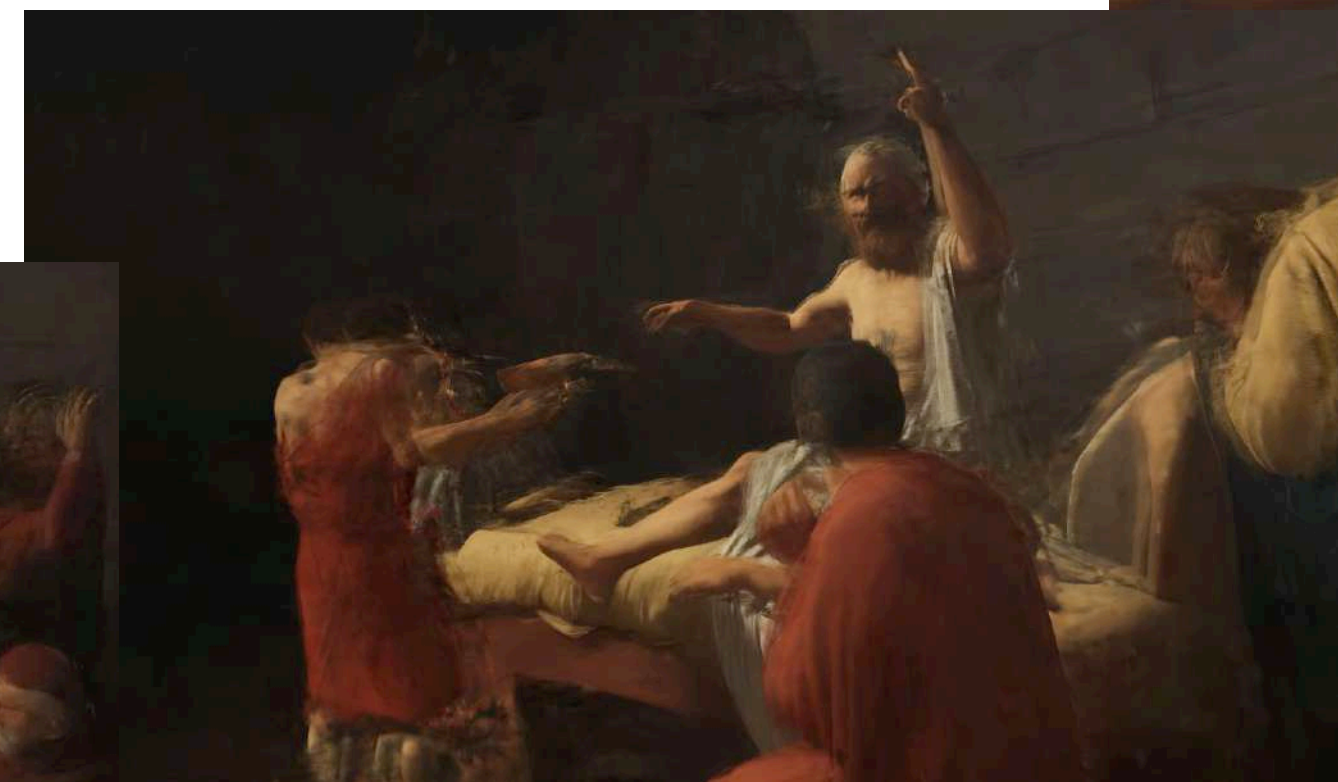
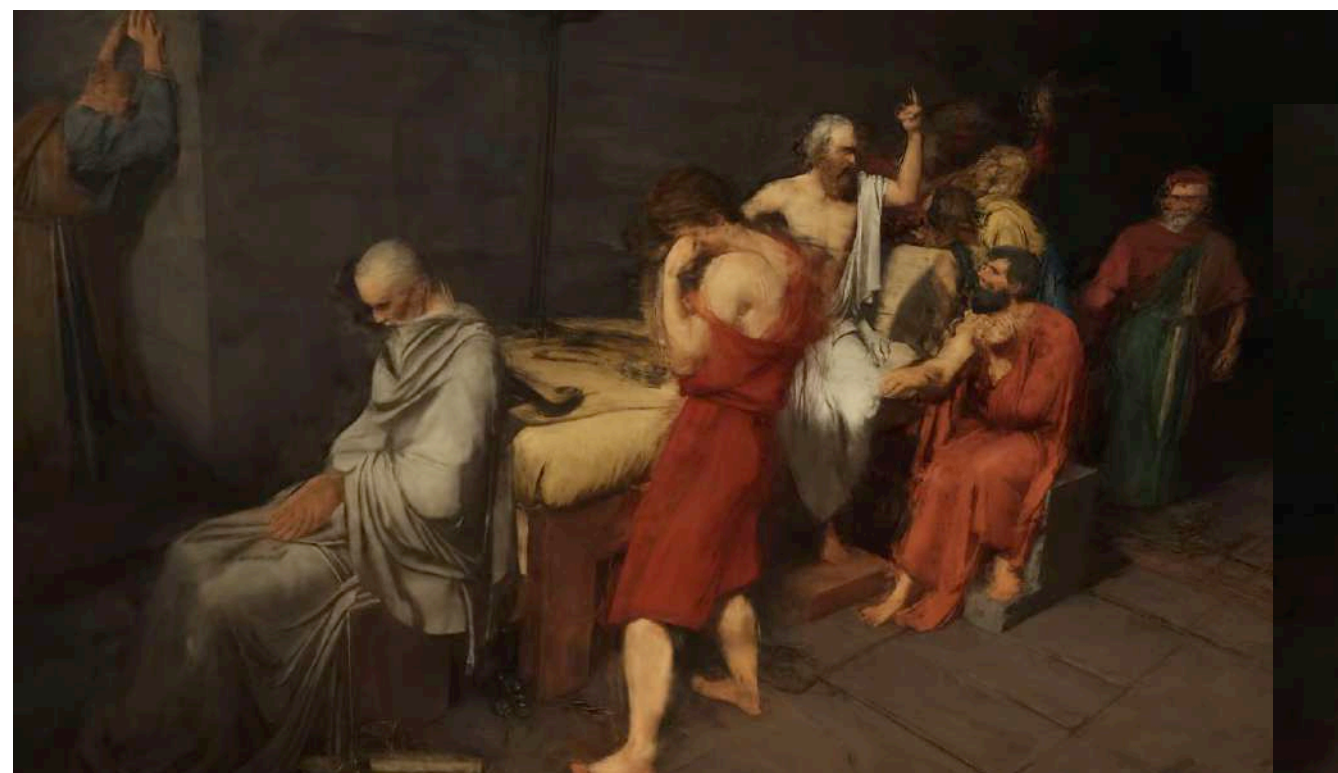
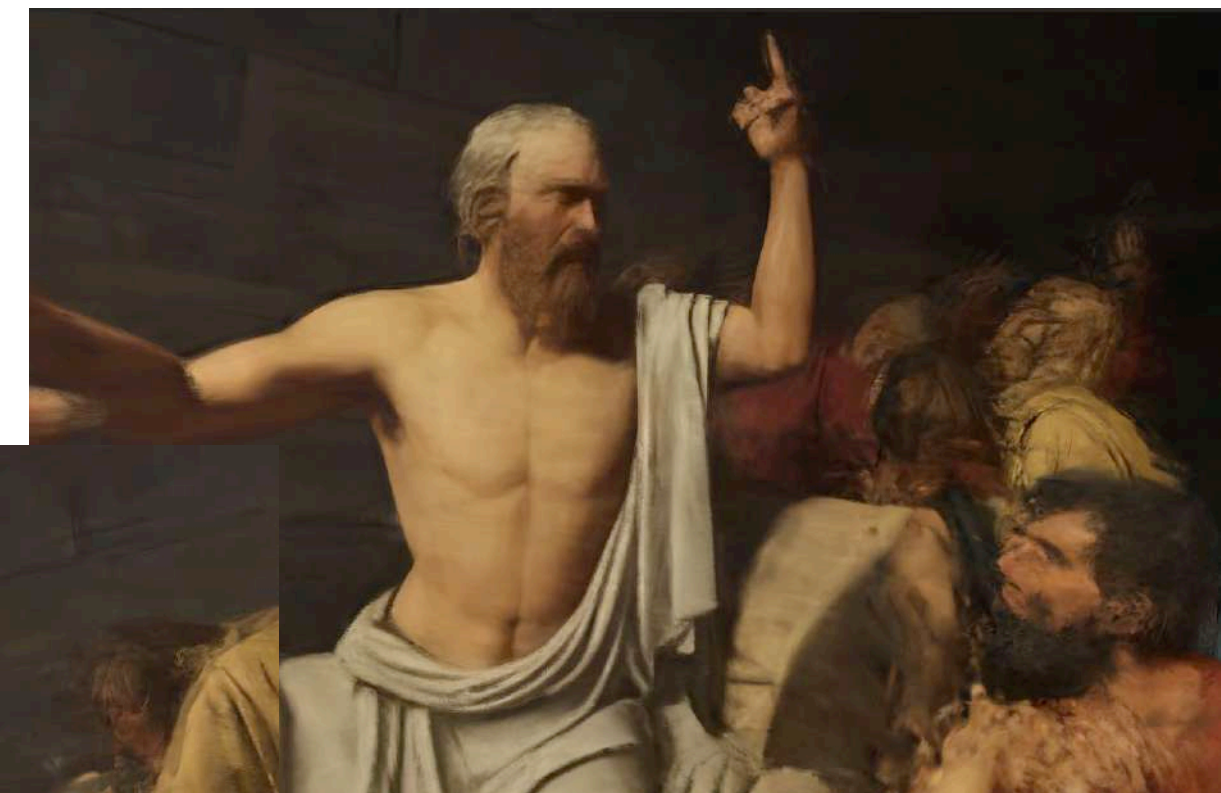
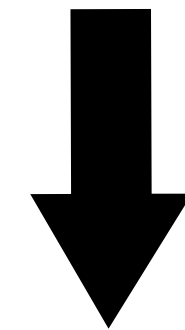


Diffusion Models in 3D

Kirill Struminsky, 2026

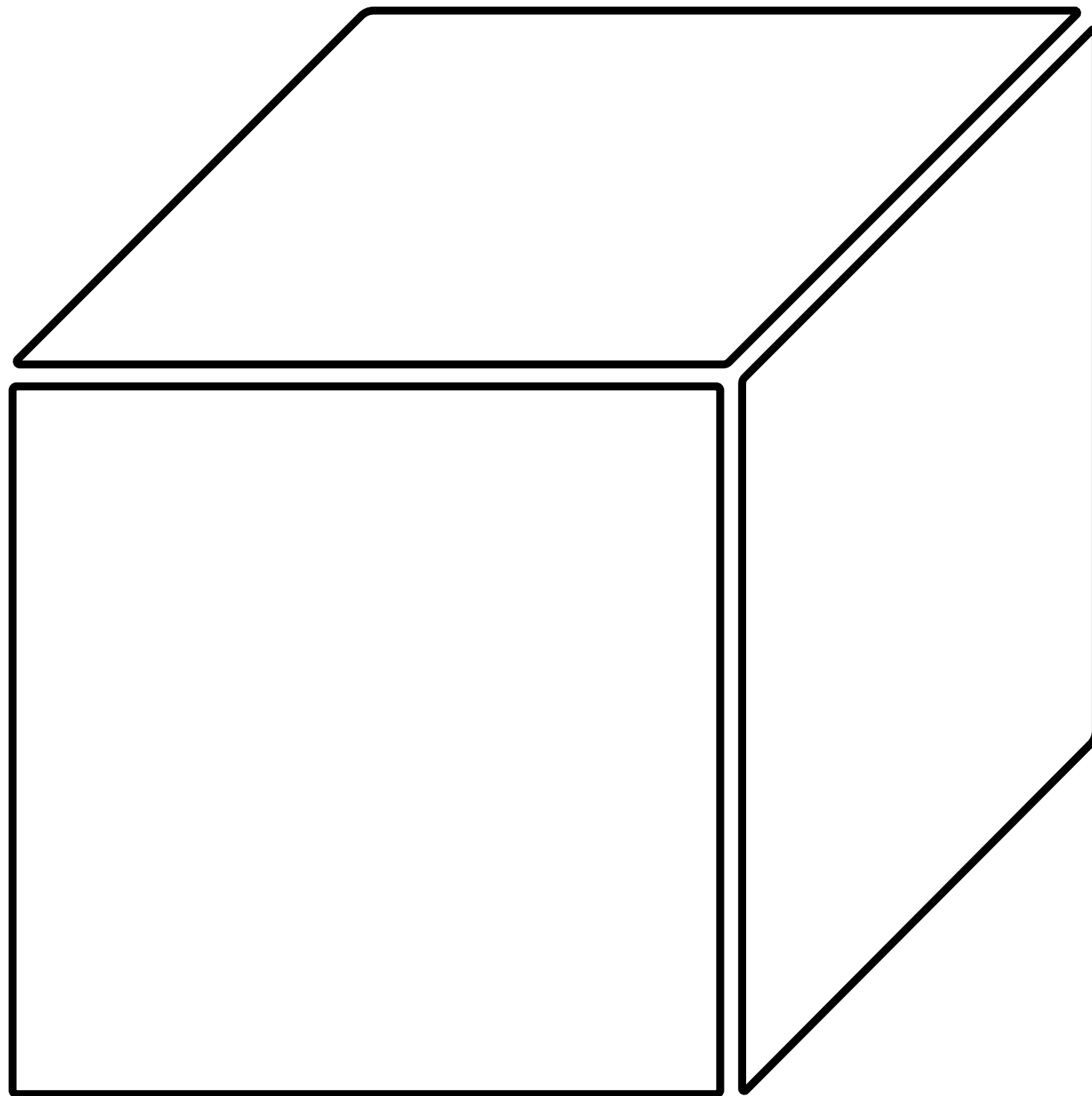
Today's Agenda

- So far in this course, generative models produced things you look at: images and videos
- Today: generating things you interact with
 - 3D objects you can rotate, inspect, 3D-print
 - Environments you can walk through
 - Worlds that respond to your actions

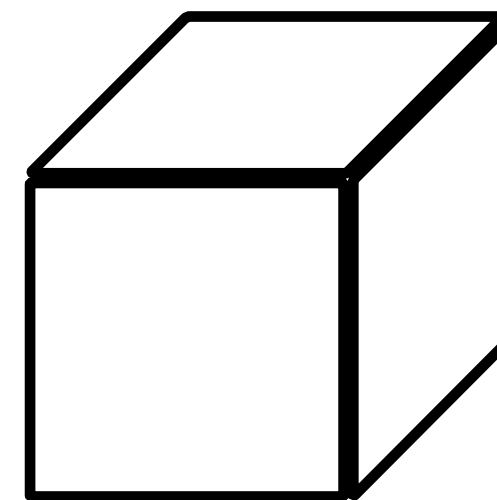


Generative Modelling in 2D, 3D, and even 4D

- A recipe for success in modern generative AI is *algorithms + compute + data*
- So far, we have discussed images & videos, what about 3D data?



LAION-5B



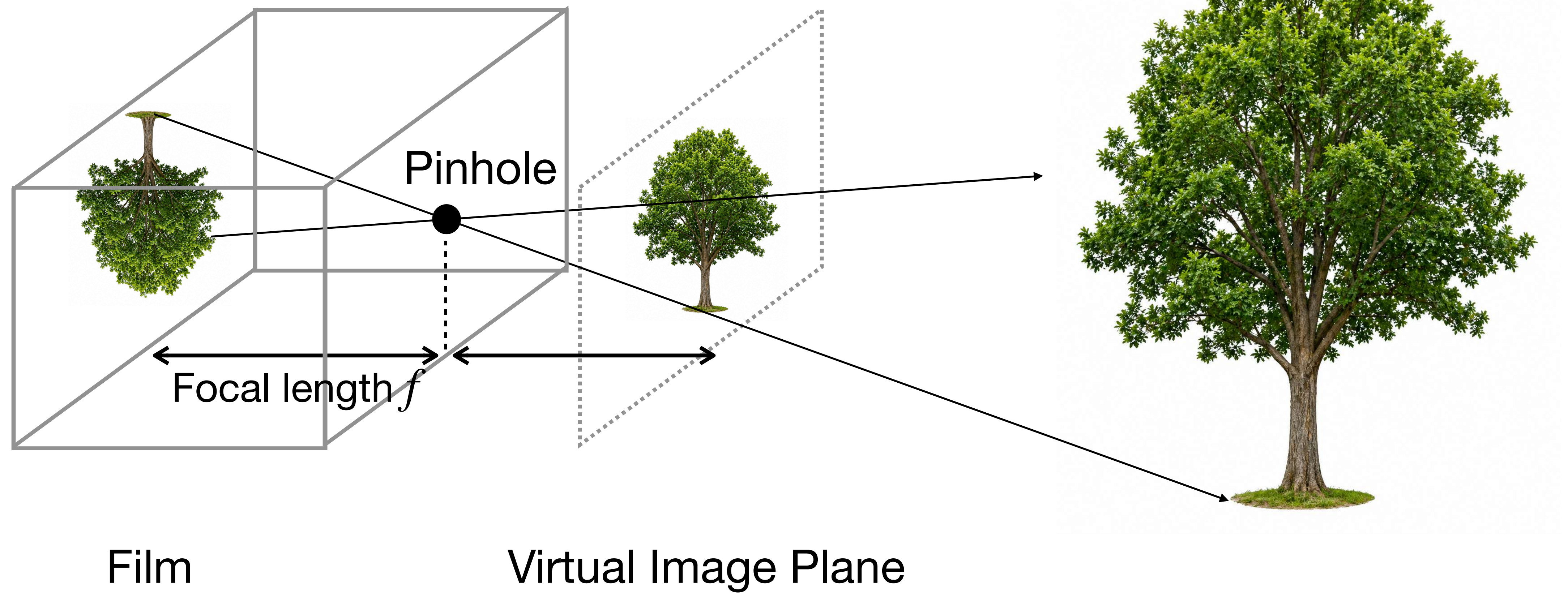
InternVid



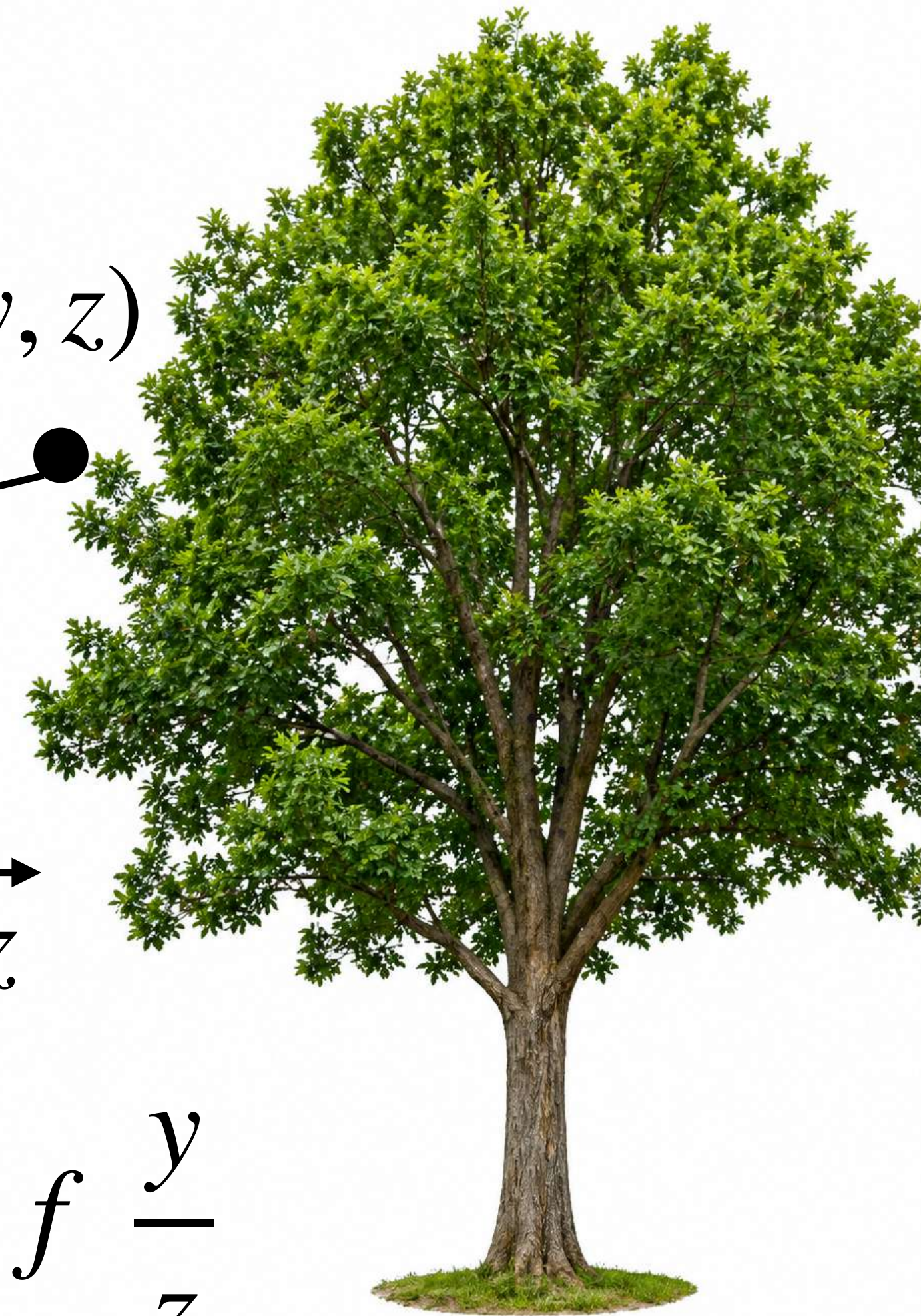
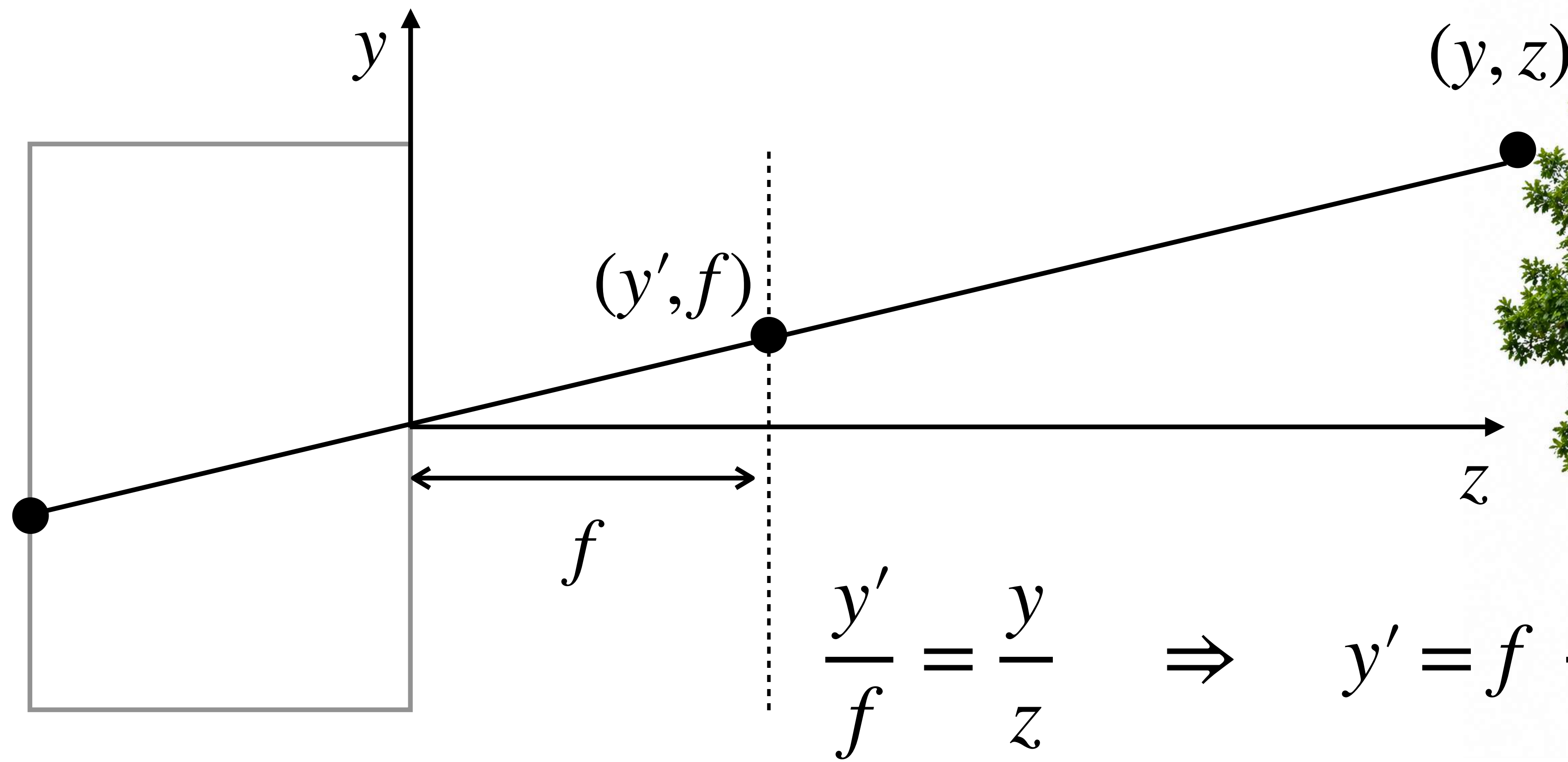
Objaverse-XL

A Primer in 3D Vision

Perspective Projection



Perspective Projection



Camera Position

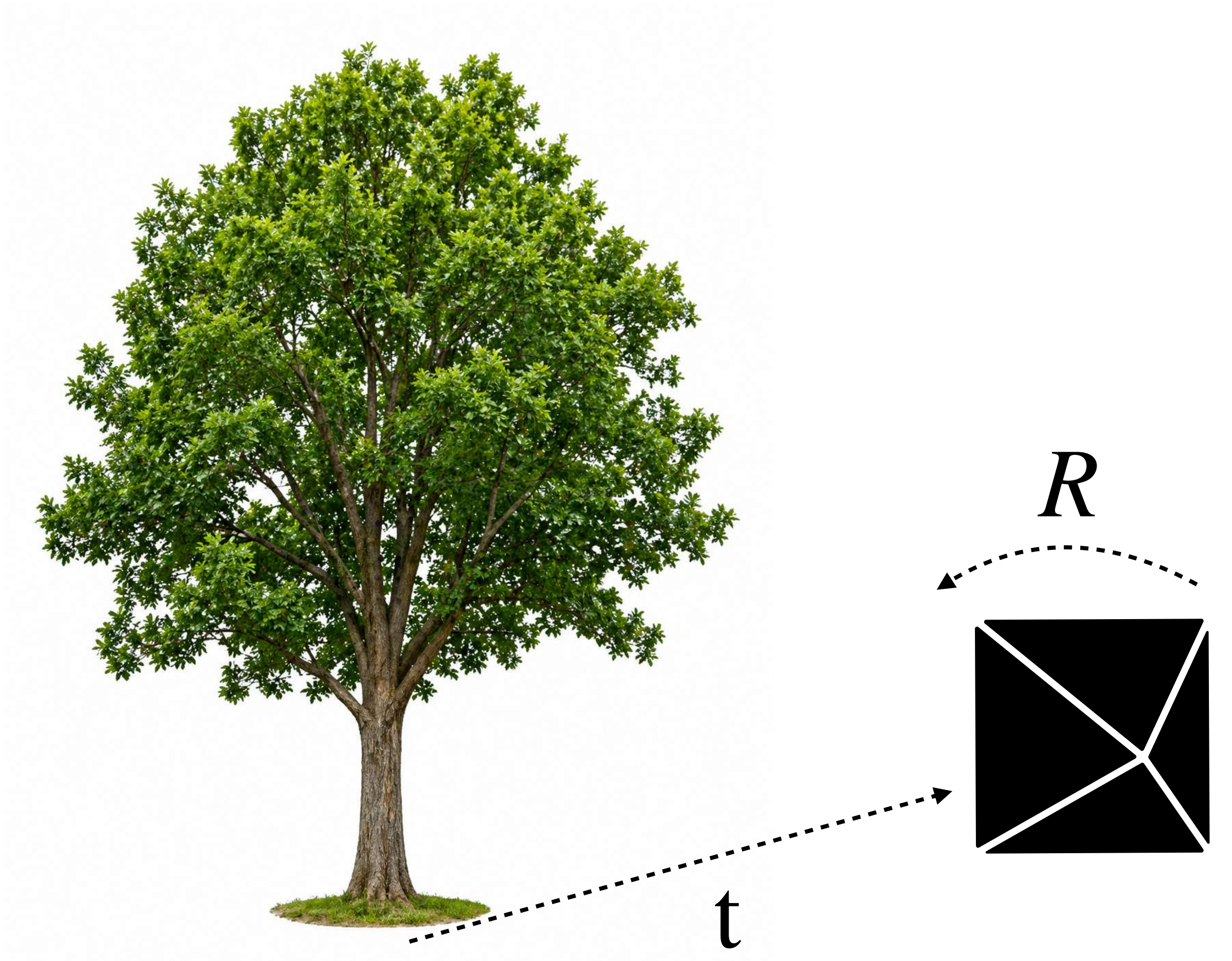
Extrinsic Parameters

- Rigid movement $y = R(x - t)$
- In homogeneous coordinates translates into

$$y = \begin{bmatrix} R & -Rt \\ 0 & 1 \end{bmatrix} x = C^{W2C} x$$

- Perspective transformation turns into

$$H = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & -Rt \\ 0 & 1 \end{bmatrix}$$



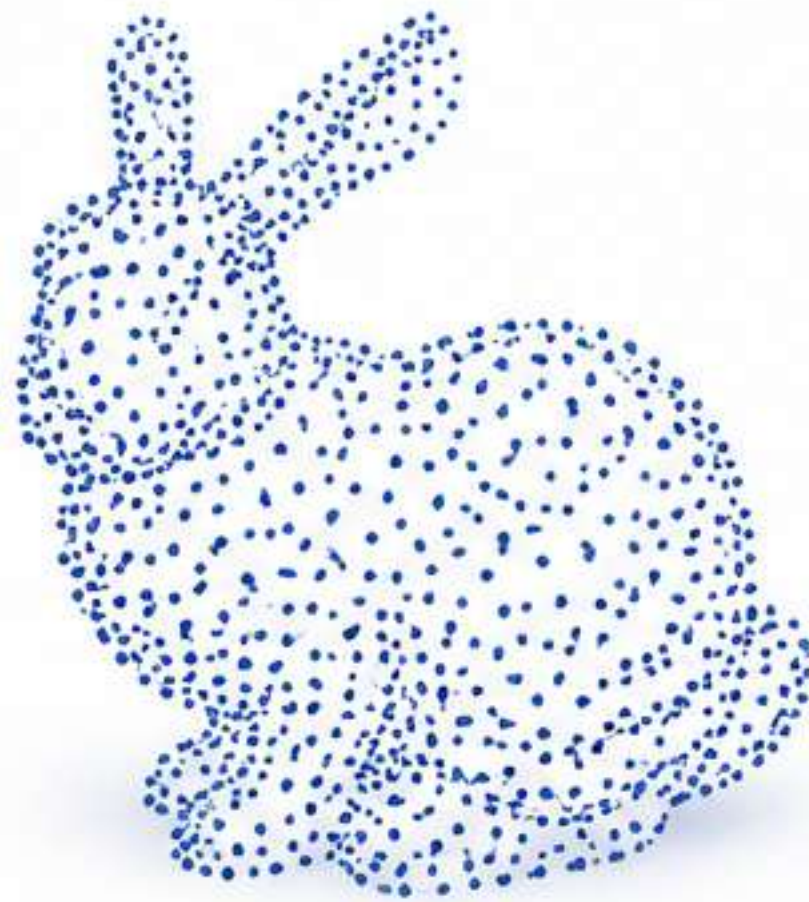
Perceiving Shapes



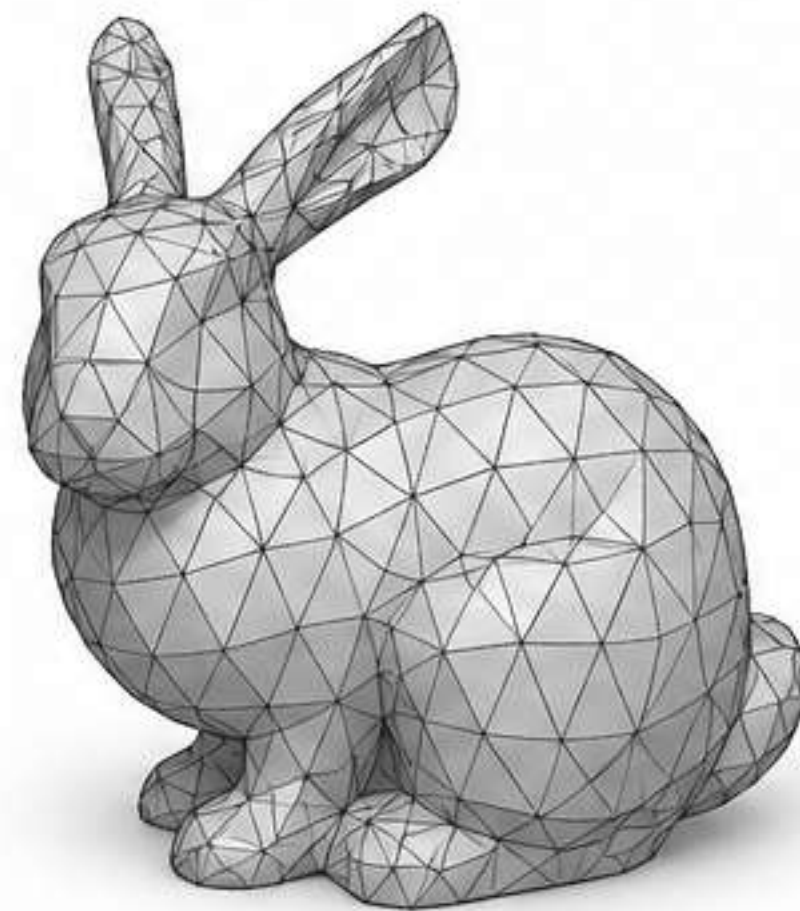
Shape Representations

- There are many ways to define a 3D shape

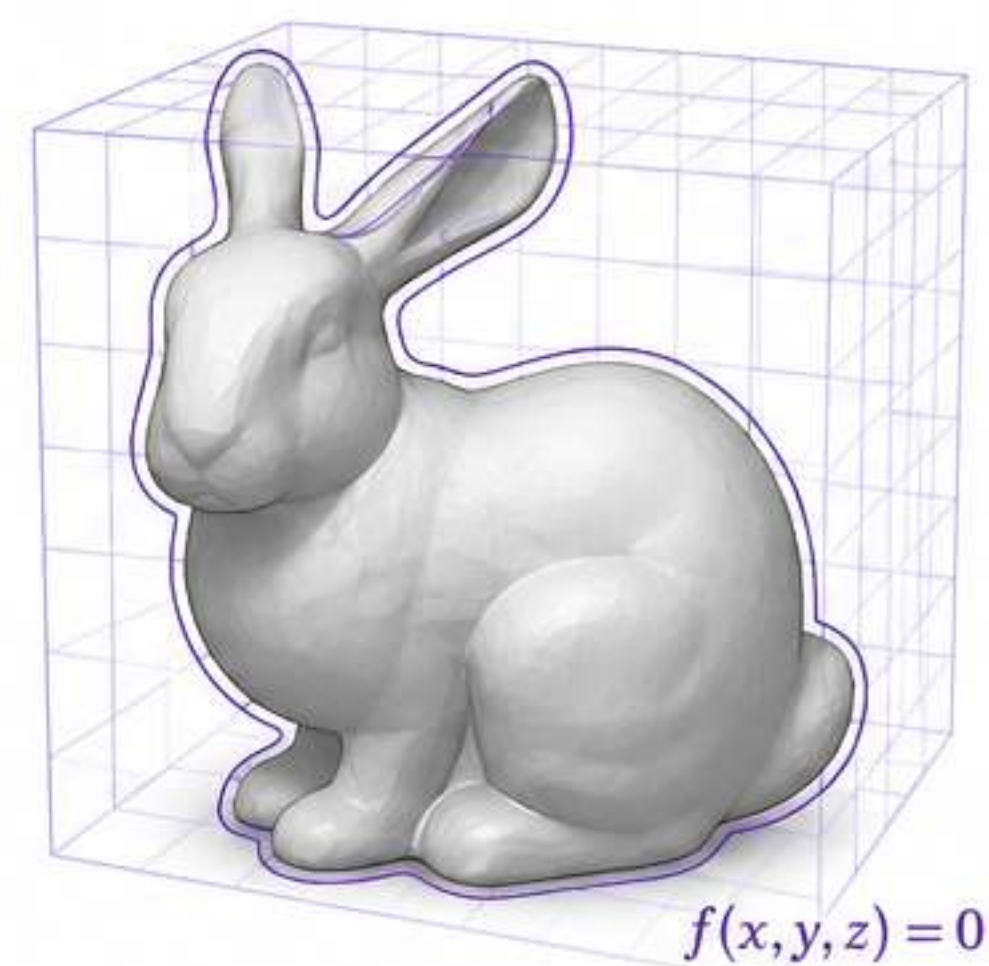
1 Point Clouds



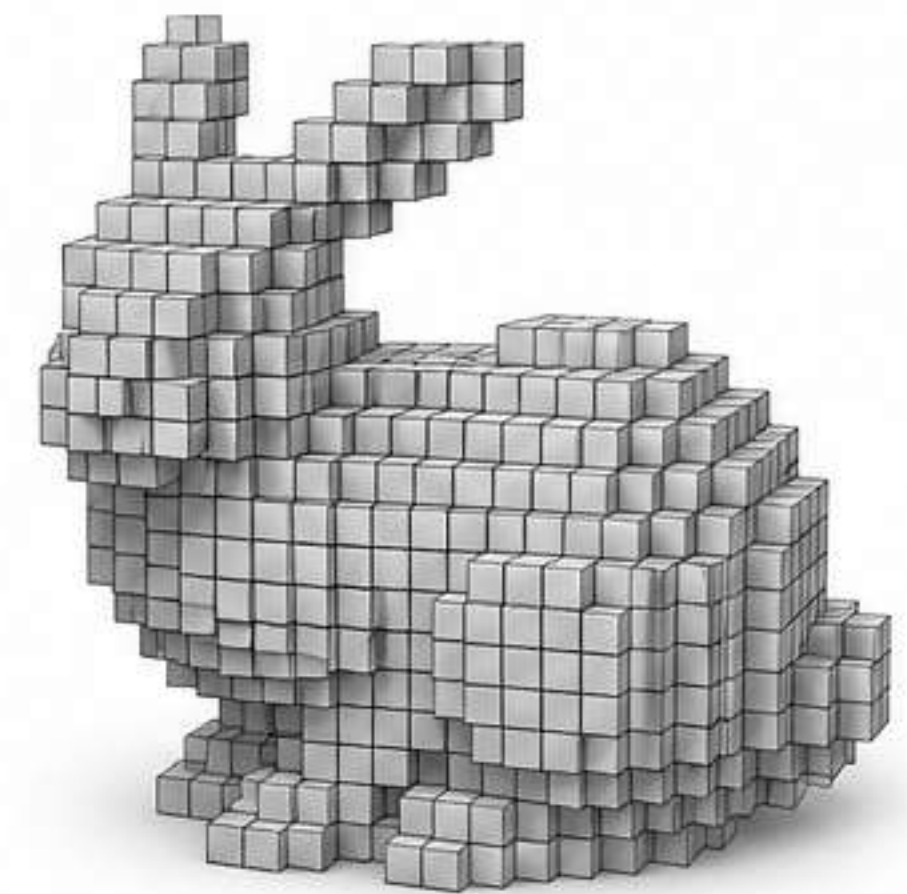
2 Polygonal Meshes



3 Implicit Surfaces



4 Volumetric Representations (Voxels)



- In practice, one chooses the representation that best suits your application
- Given cameras, shape can be recovered from images via gradient descent (NeRFs, Splatting)

From Text-to-2d to Text-to-3D

DreamFusion

- Pre-trained text-to-2D diffusion model
- Input: text prompt
 - “A DSLR photo of a frog wearing a sweater”
- Output: 3D model
- Parameterise 3D model with NeRF $x(\theta)$
- Without ground truth images, what would be the objective?



Score Distillation Sampling (SDS)

- Render a NeRF based image $x_0(\theta)$ and corrupt it with noise $x_t = \alpha_t x_0 + \sigma_t \epsilon$

- Estimate the uncorrupted image with the diffusion model $x_\phi(x_t, t)$

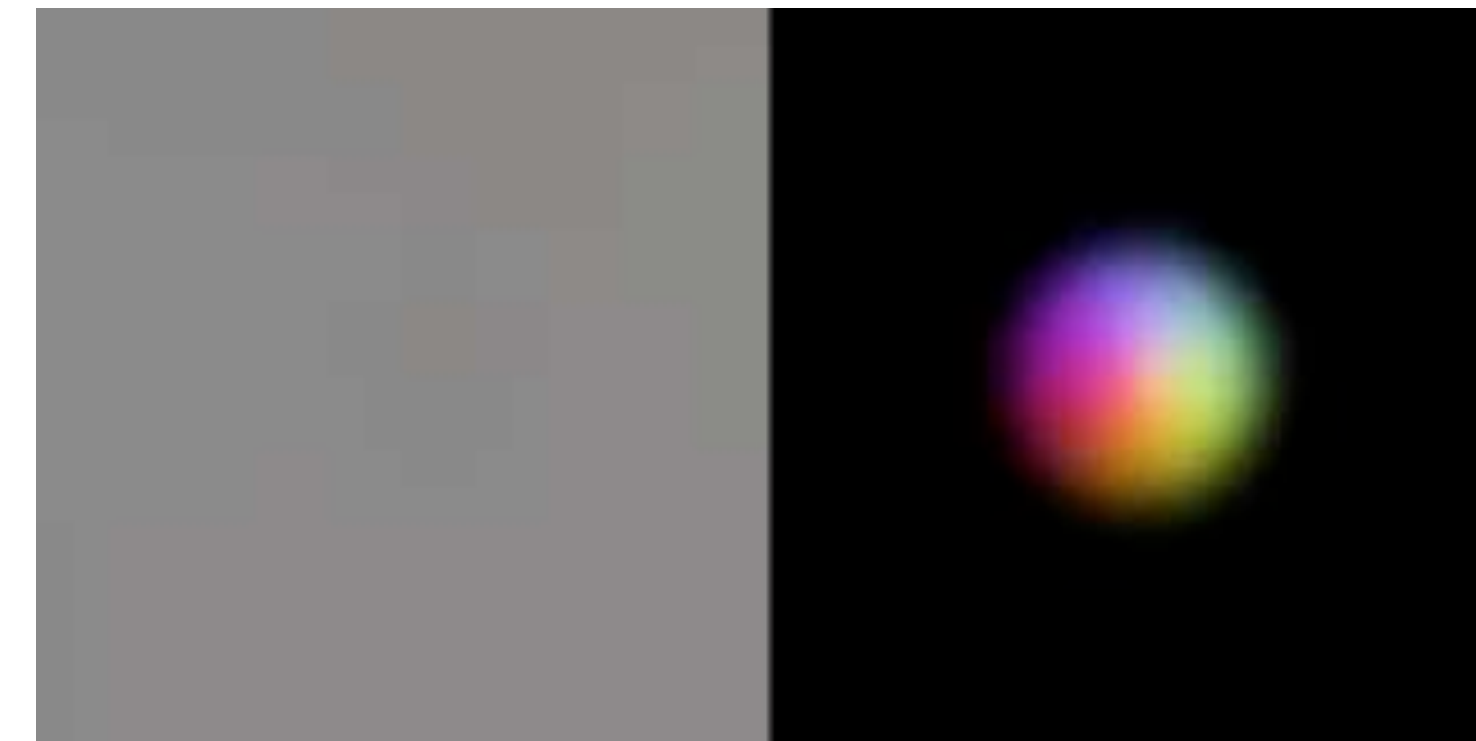
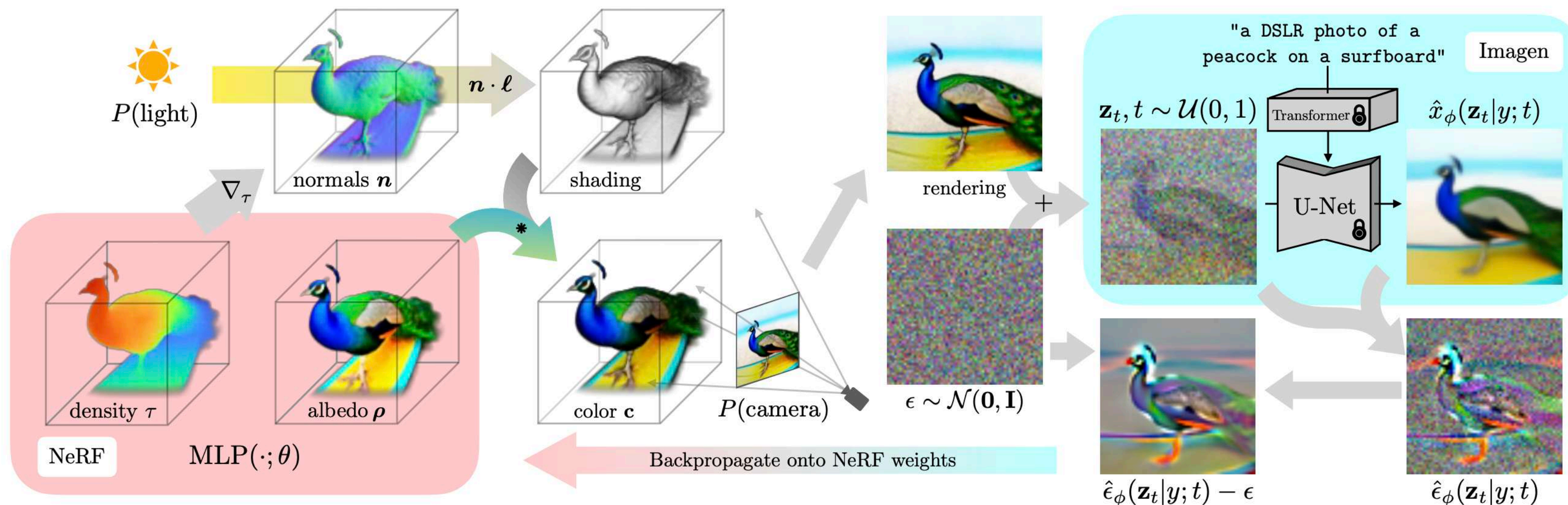
- SDS gradient:
$$\nabla_\theta \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\epsilon} \left[w(t) \left(\frac{\partial x_0}{\partial \theta} \right)^\top (x_0(\theta) - x_\phi(x_t, t)) \right]$$

 Surrogate:
$$= \nabla_\theta \mathbb{E}_{t,\epsilon} \left[\frac{w(t)}{2} \|x_0(\theta) - \text{sg}[x_\phi(x_t, t)]\|_2^2 \right].$$

sg: stop gradient

w(t): positive weight; includes noise-to-image scaling.

- ADD combines score distillation with an adversarial loss.



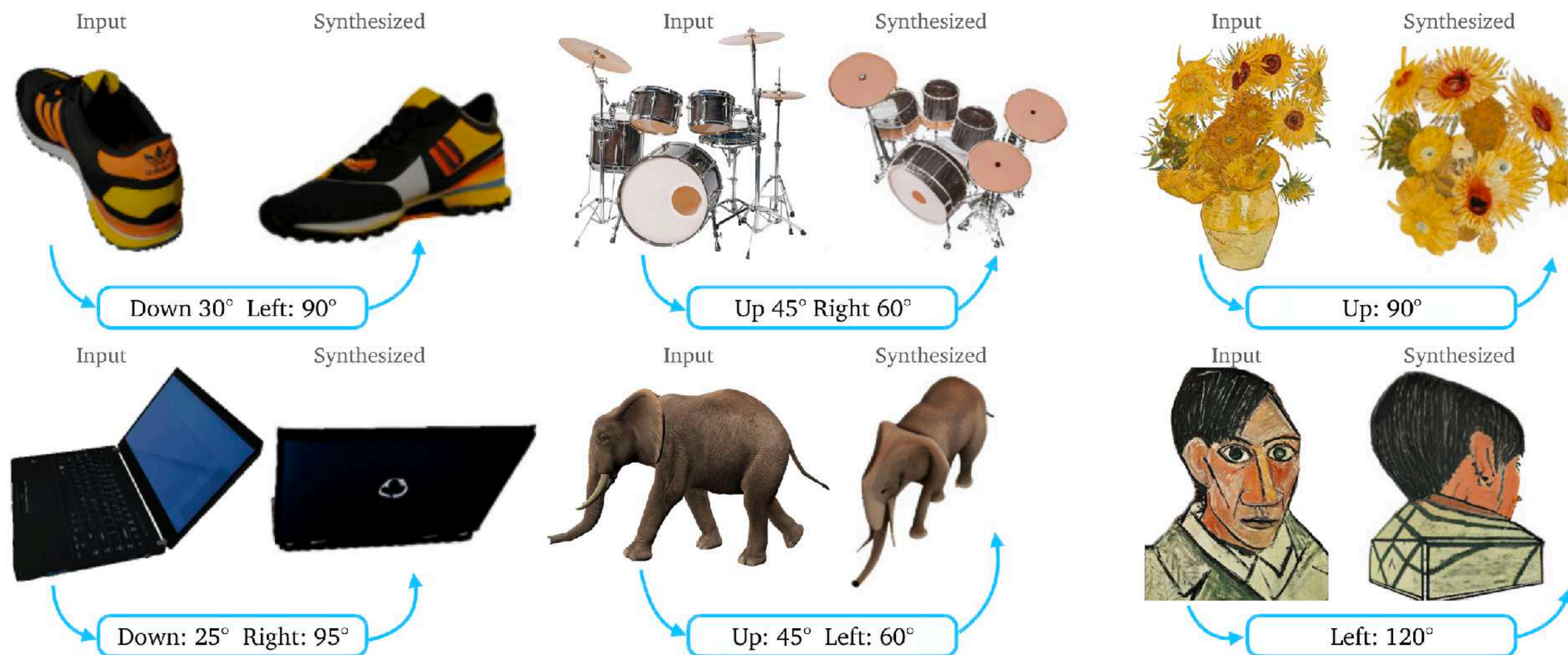
DreamFusion: Results

- Pros:
 - It works!
- Cons:
 - ... and sometimes it does not
 - Low sample diversity due to mode-seeking objective
 - Janus problem: 2D model tries to treat all sides of the model equally



Camera-conditioned Models

- Diffusion models excel at conditional generation
- Zero-1-to-3: we can fine-tune a 2D model to predict novel views in 3D on synthetic data



Distilling Camera-Conditioned Models in 3D

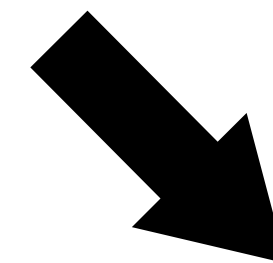
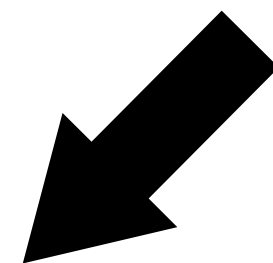
- Novel-views can be inconsistent across samples
- We can use the camera-conditioned model as a backbone for SDS and distill generated images in a consistent 3D model



The Fork

- DreamFusion showed that 2D diffusion knows a lot about 3D, but SDS alone is fragile, slow, and limited
- As a result, the field split into two research directions

DreamFusion



Train dedicated 3D generative models to generate assets directly

Move away from 2D diffusion, train on large-scale synthetic 3D data

Use diffusion as an engine for novel view synthesis to reconstruct real scenes

Keep 2D diffusion, fine-tune on real multi-view data, generate new views instead of SDS

3D Generation on Synthetic Data

From Image-Based Models to Native 3D Diffusions

Objaverse & Objaverse-XL

- Contain 800k & 10m+ 3D assets correspondingly
- Quality varies widely; most pipelines filter down to 300–500K usable assets
- Sufficient to train dedicated 3D generation models for the first time
- Synthetic data leads to synthetic appearance

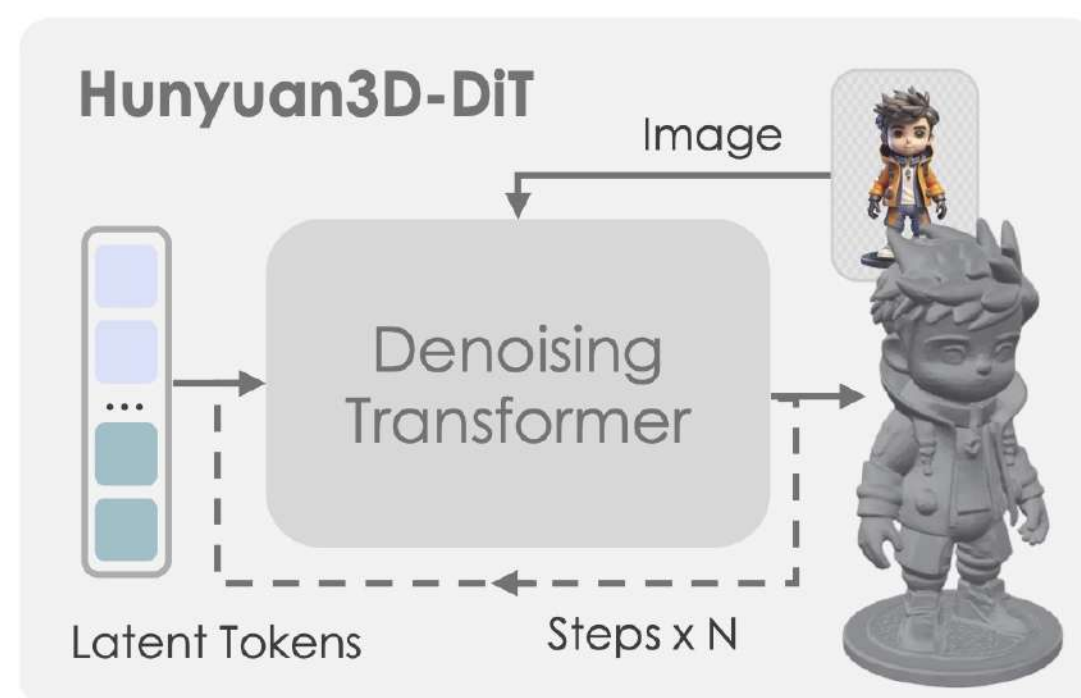
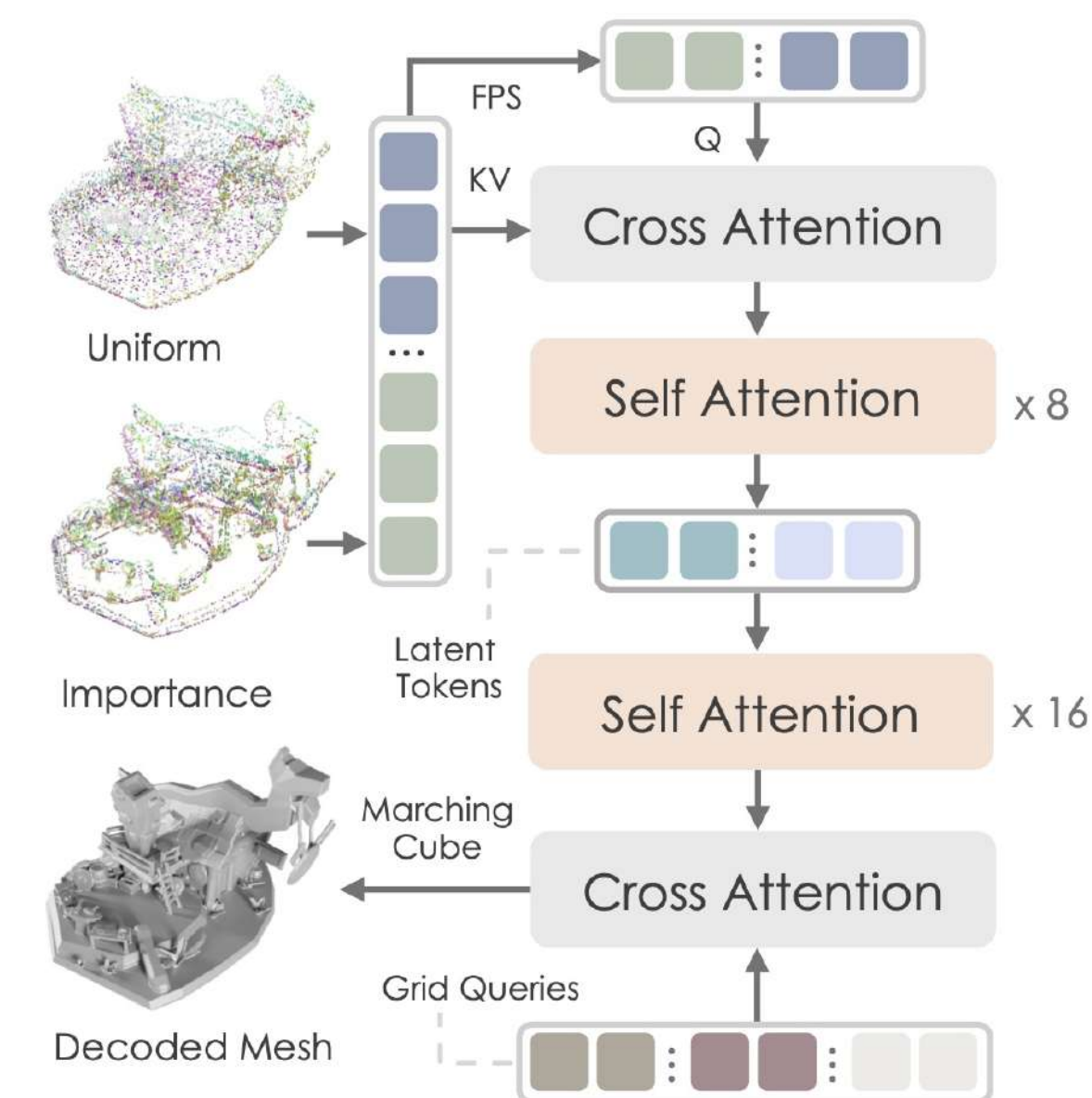


Hunyuan3D (1/2)

- Split task into shape & texture generation
- Dedicated latent diffusion shape model
- Texture generation based on multi-view diffusion

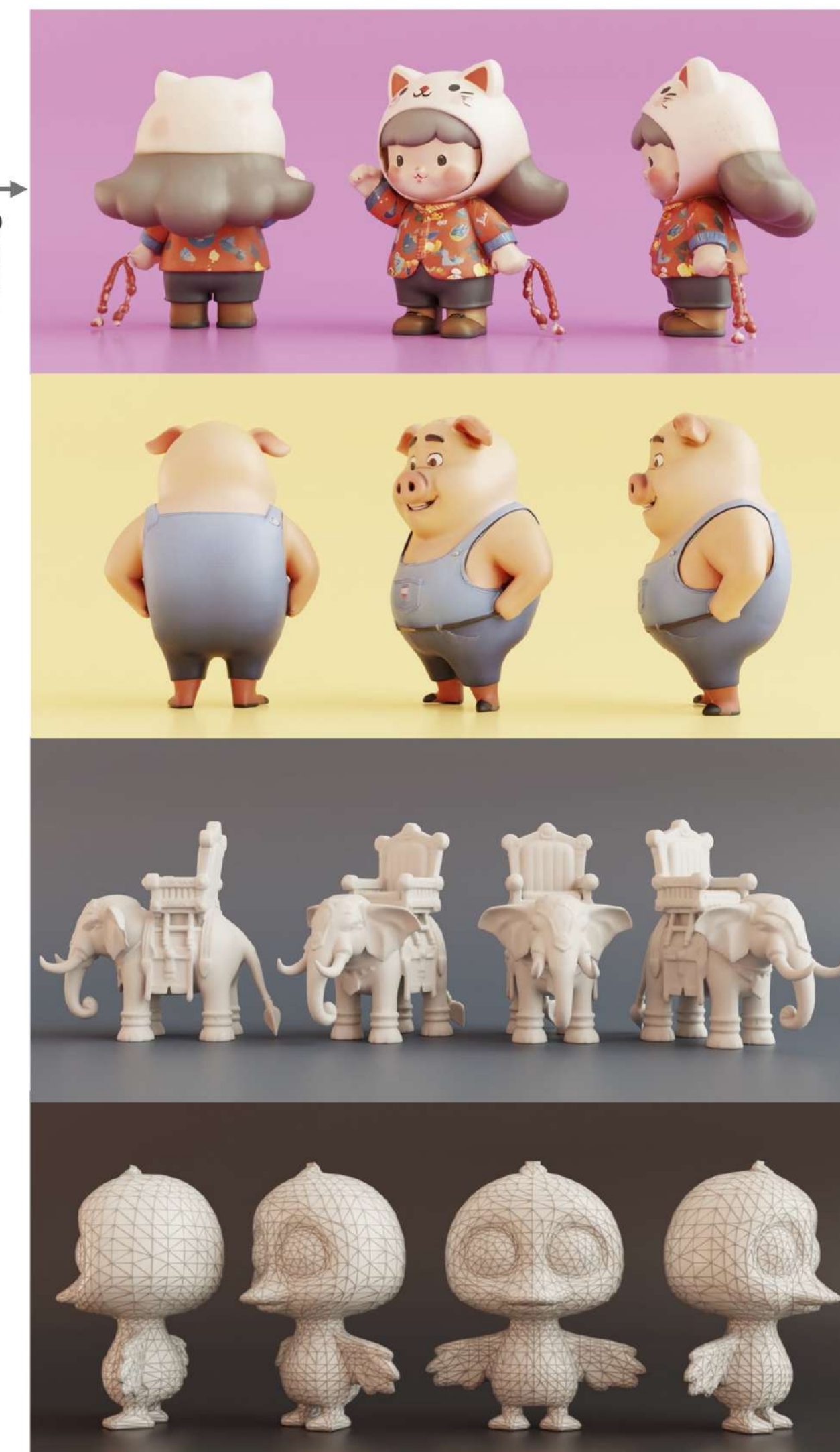
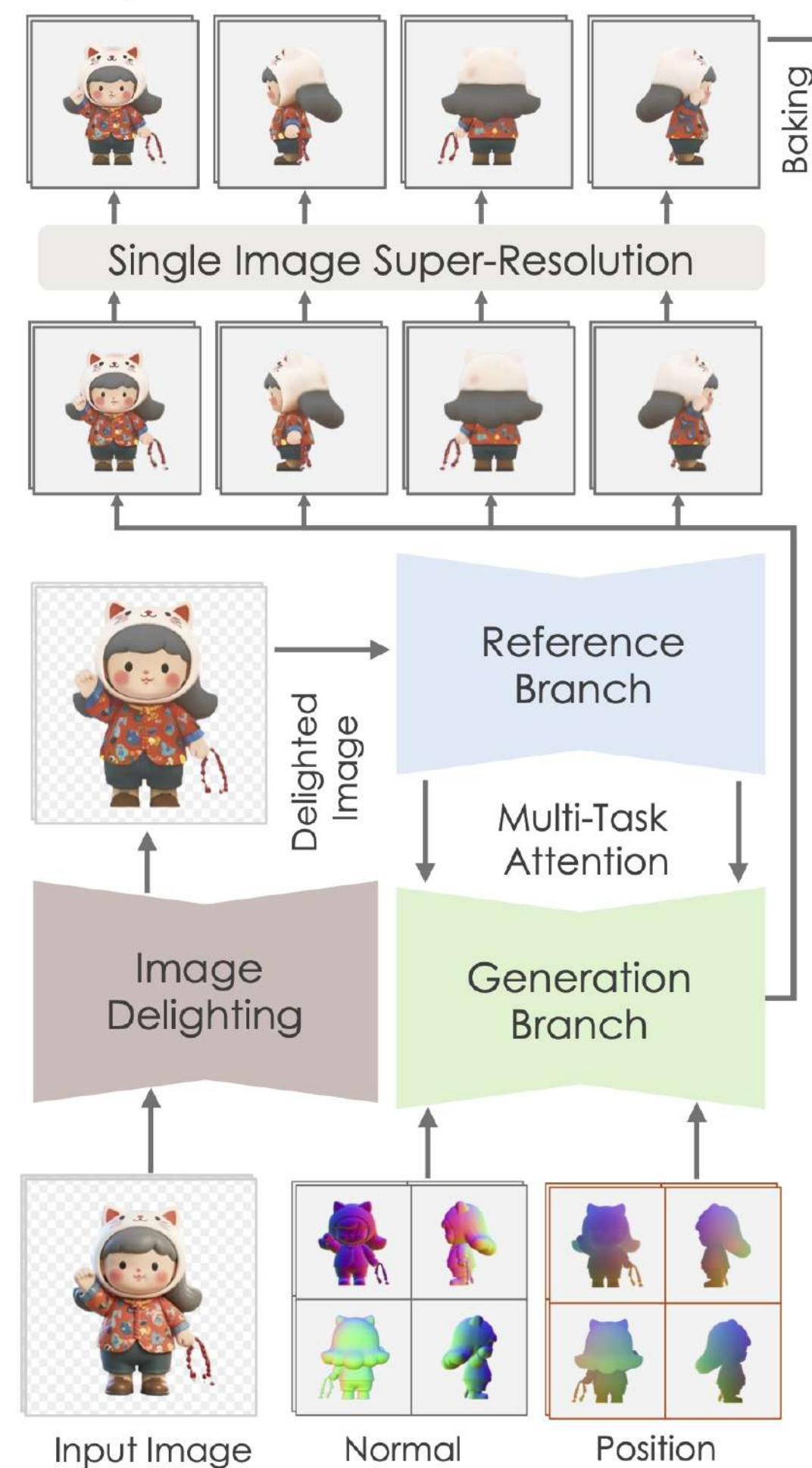
Shape Generation

Hunyuan3D-ShapeVAE



Texture Synthesis

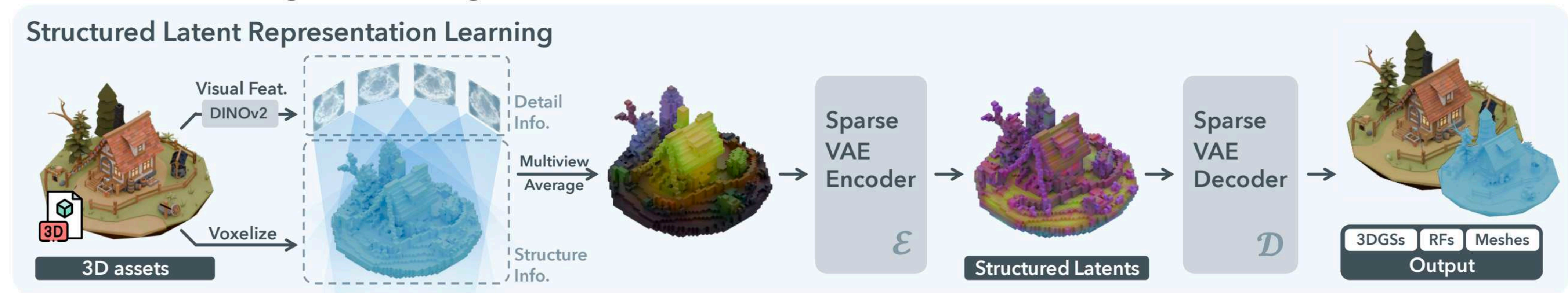
Hunyuan3D-Paint



Trellis (1/2)

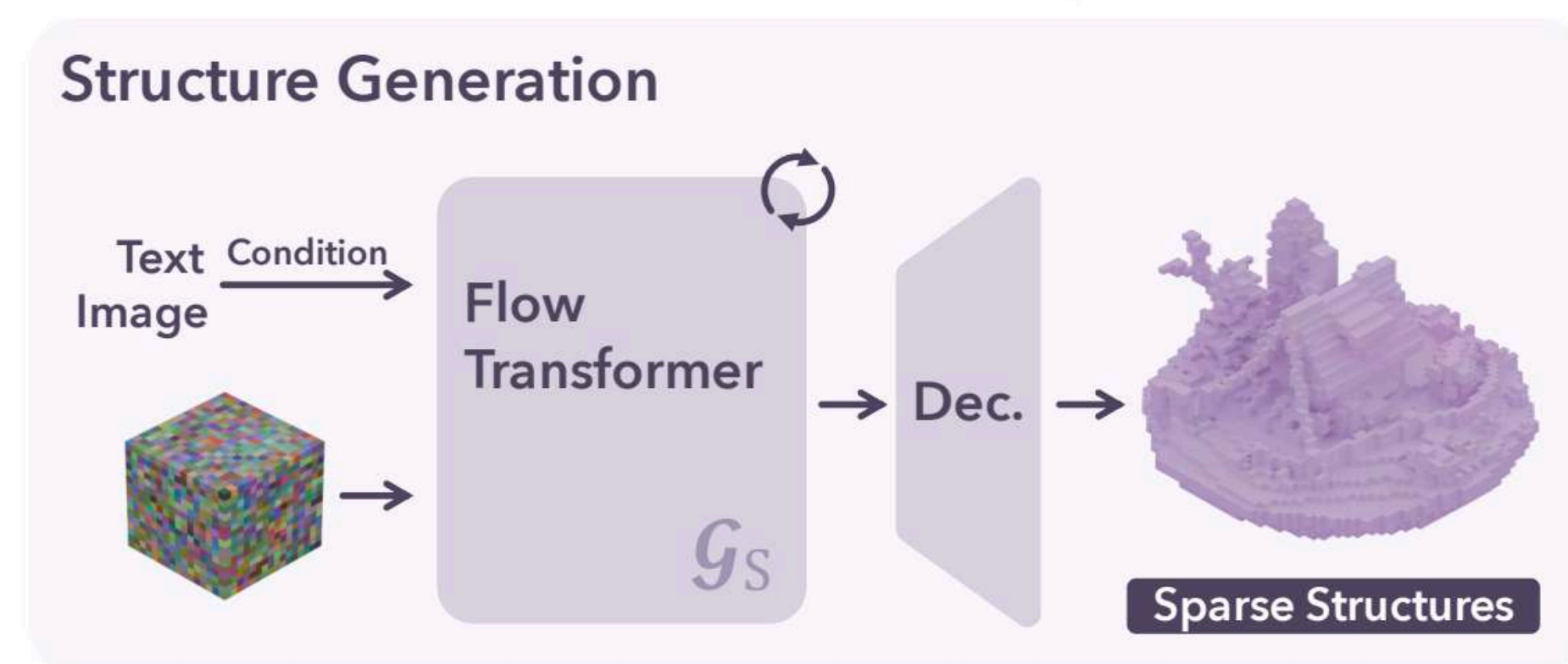
3D Assets Encoding & Decoding

Structured Latent Representation Learning

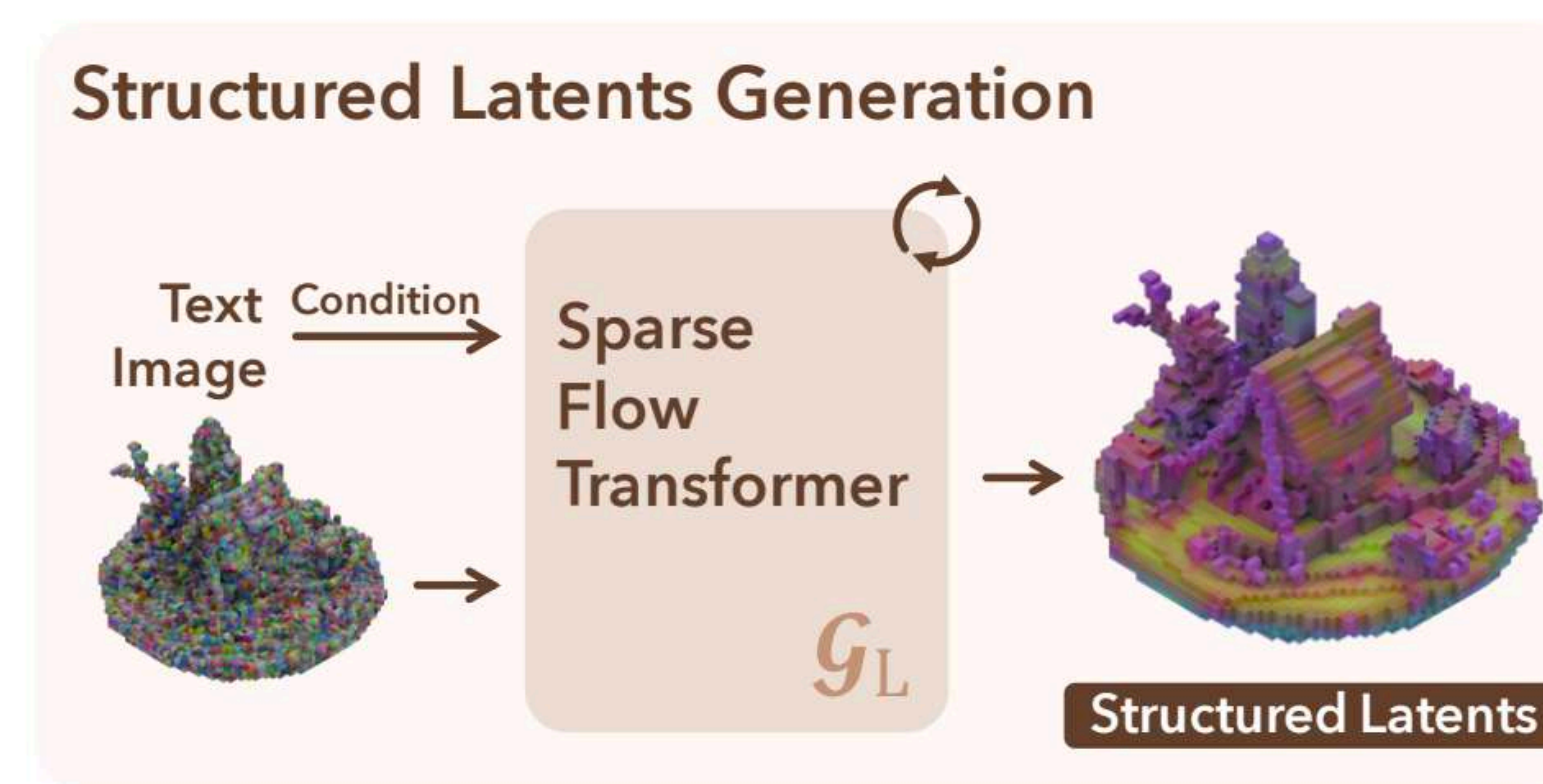


3D Assets Generation

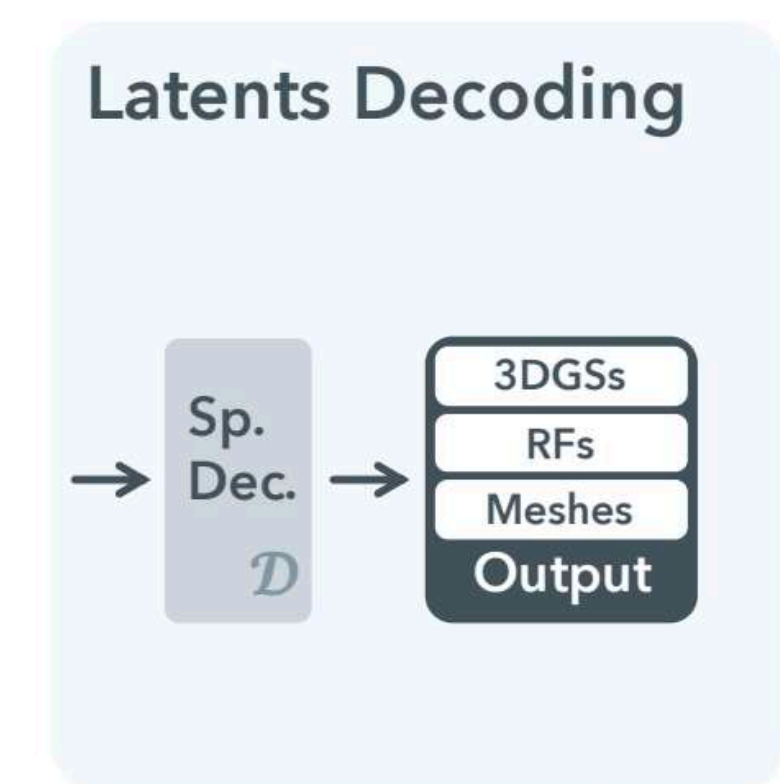
Structure Generation



Structured Latents Generation

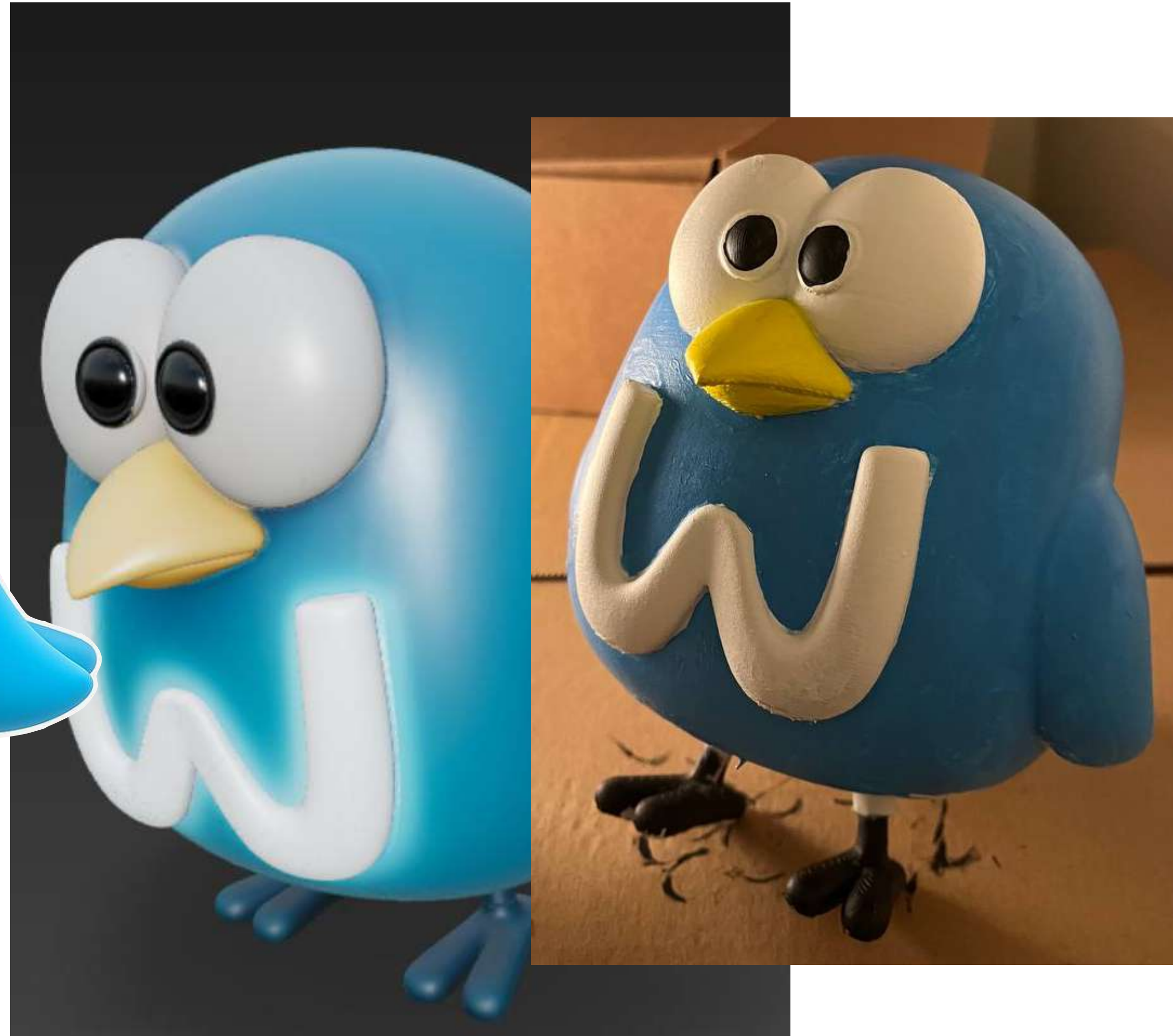
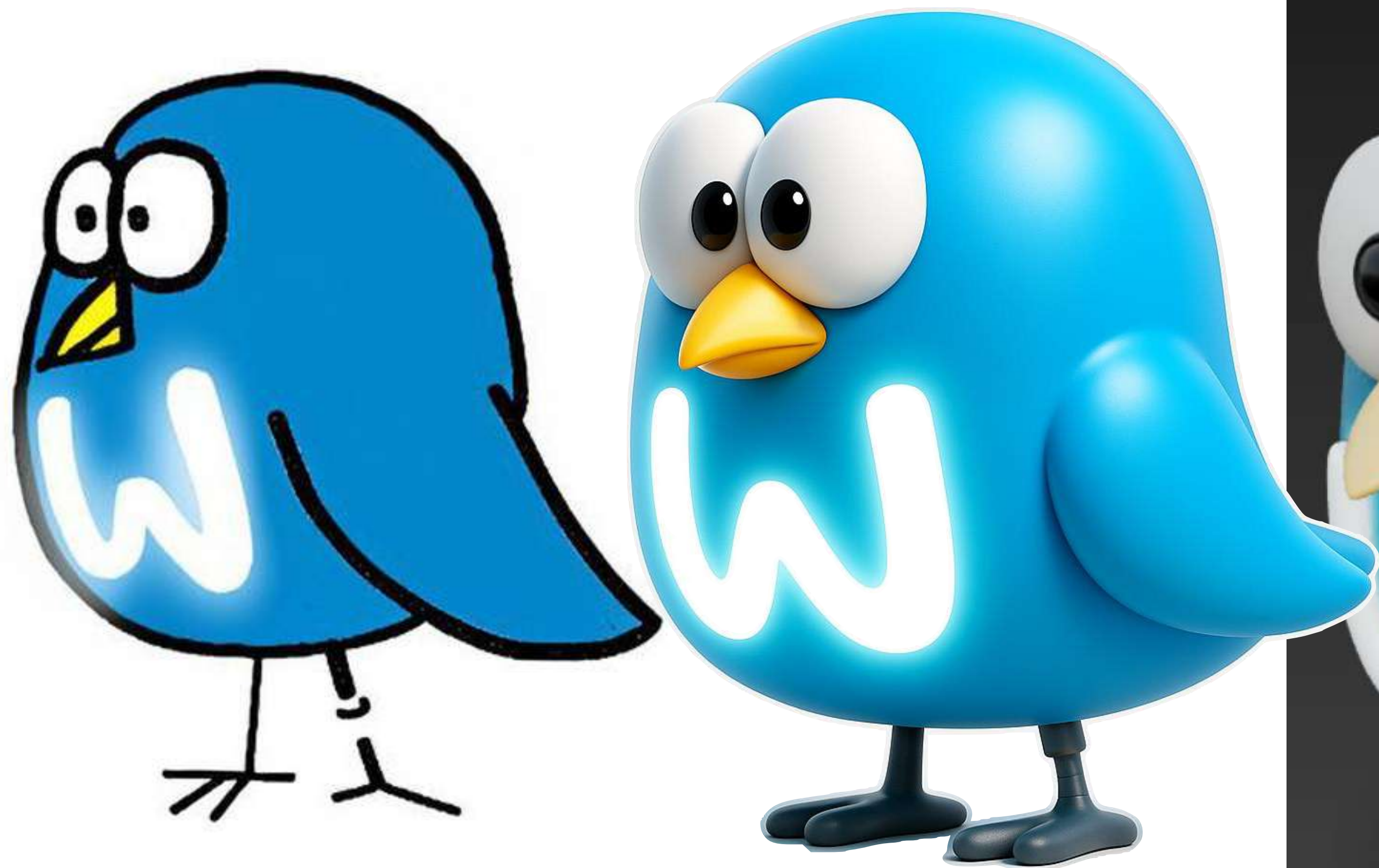


Latents Decoding



Real Use-Case

A Gift for a Friend

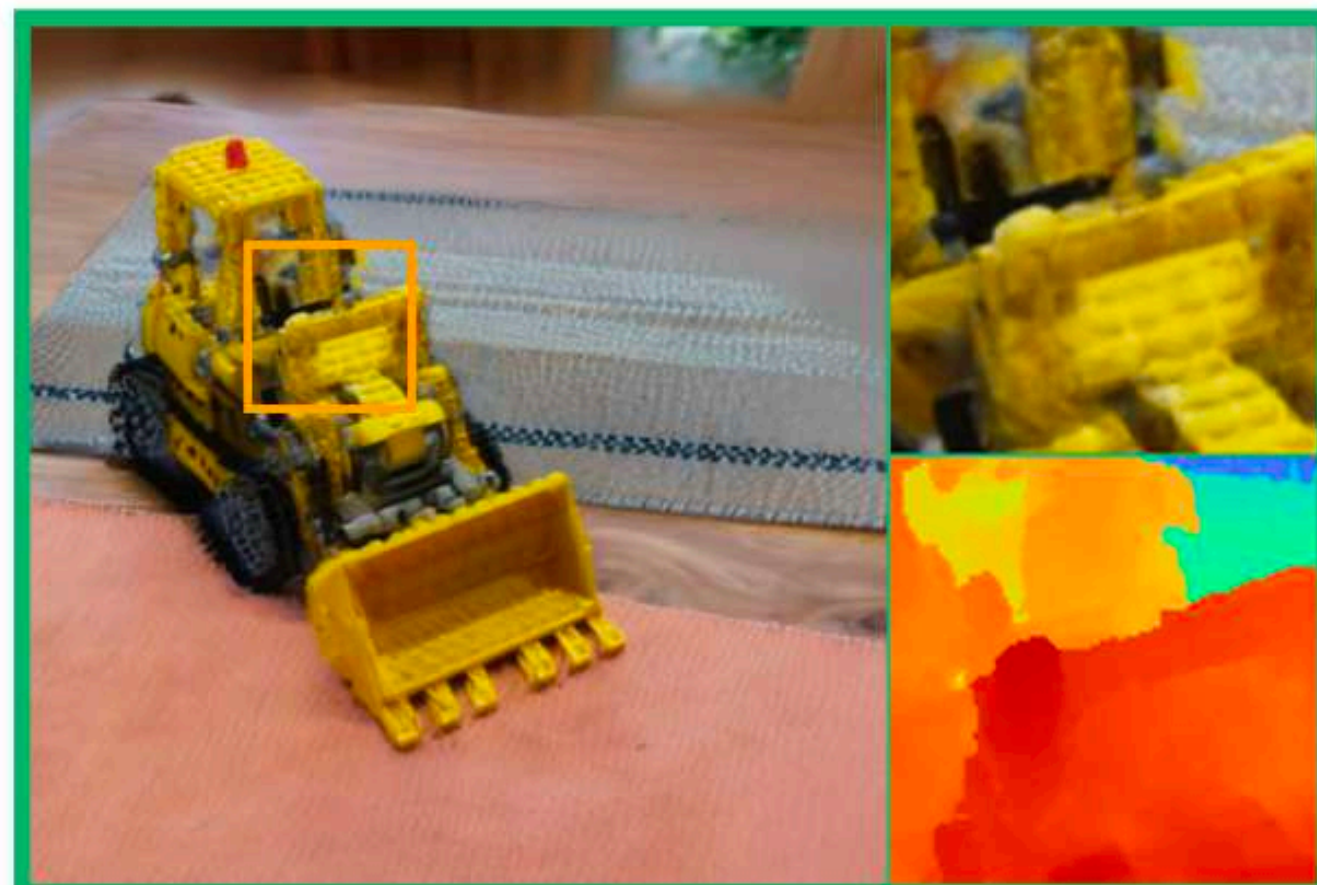


Fine-Tuning on Real Data

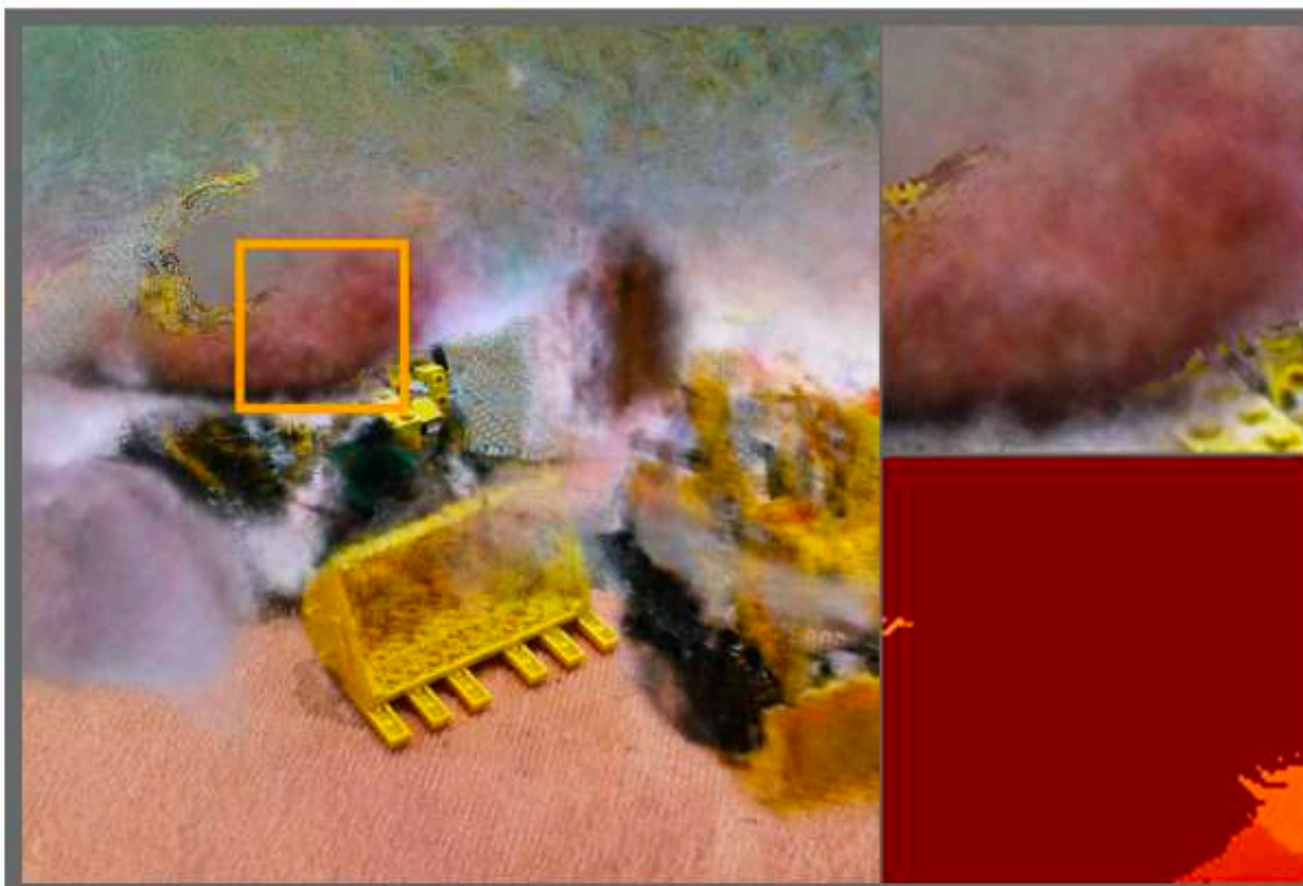
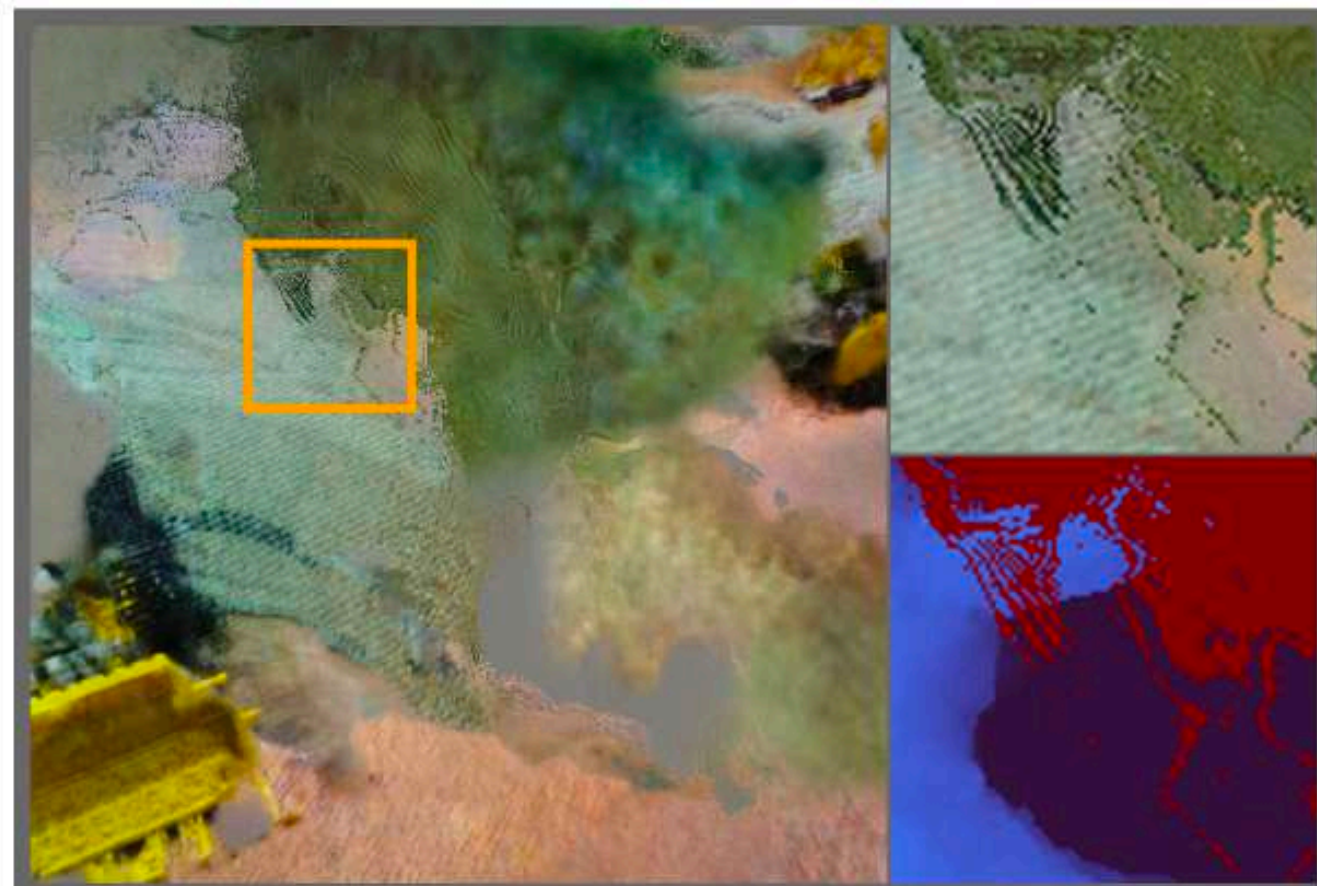


The Shift to Real Data

ReconFusion



Zip-NeRF [2]



3 views

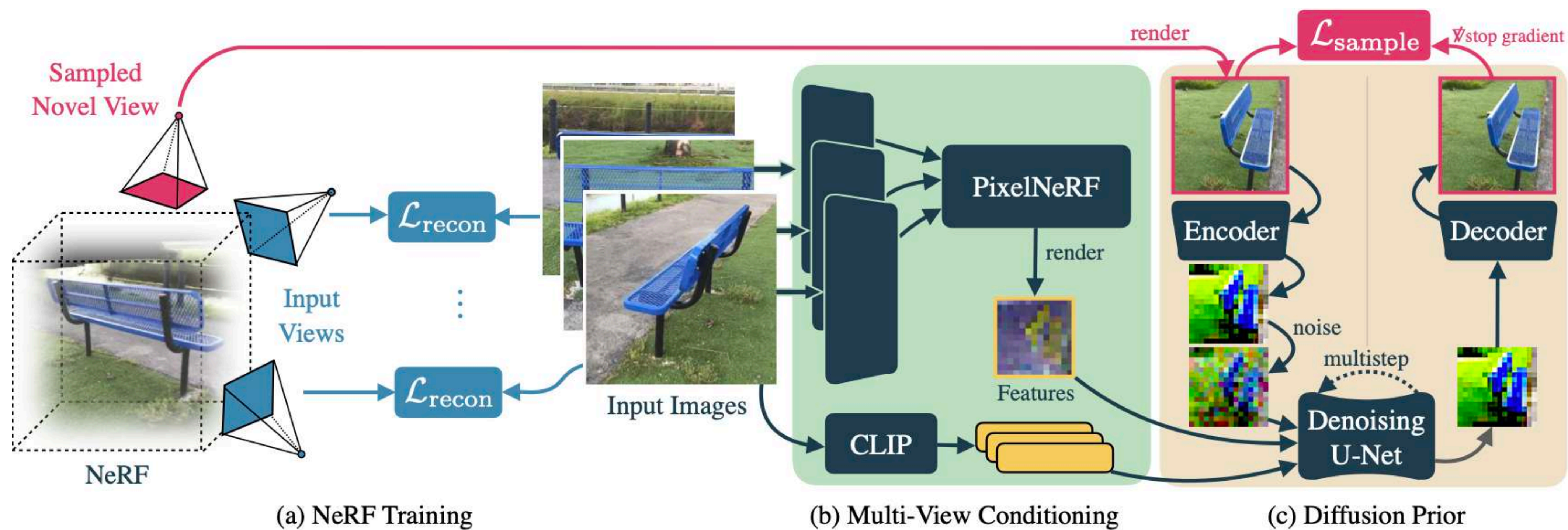
9 views

54 views

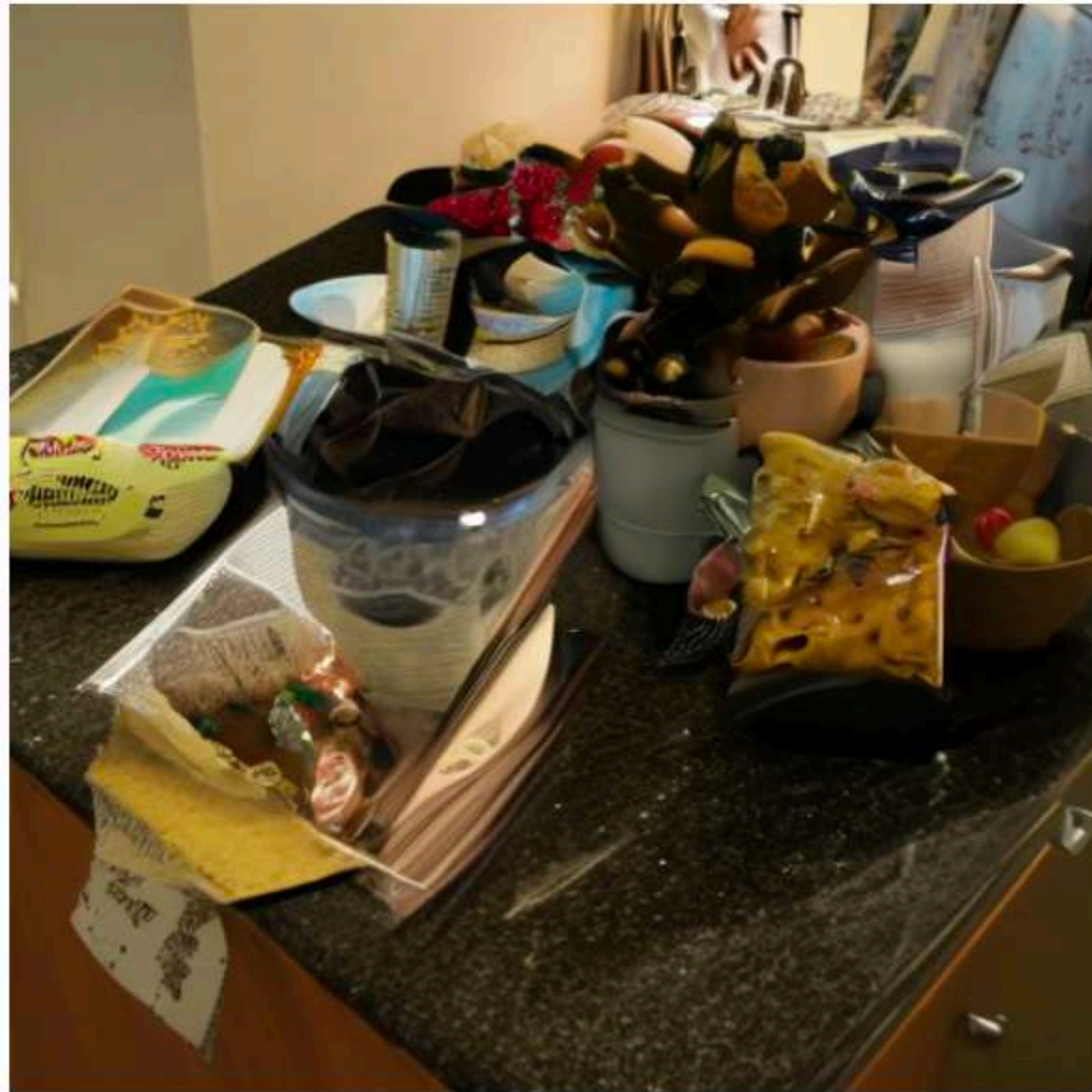
Training

- Initialise with a pretrained text-to-image model
- Fine-tune *collect as much data as you can*
 - Objaverse
 - CO3D
 - RealEstate10k
 - MVImageNet

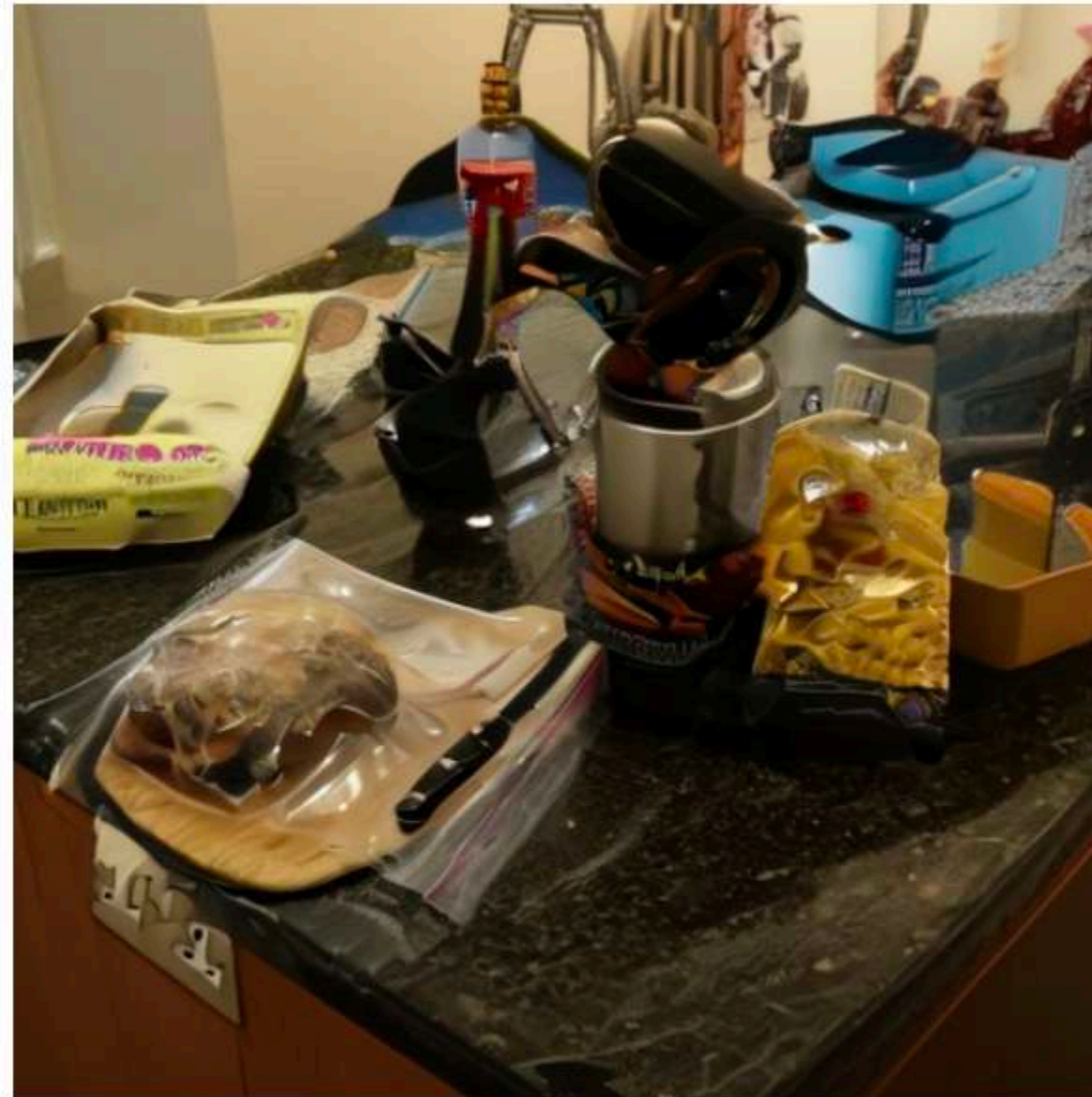
ReconFusion



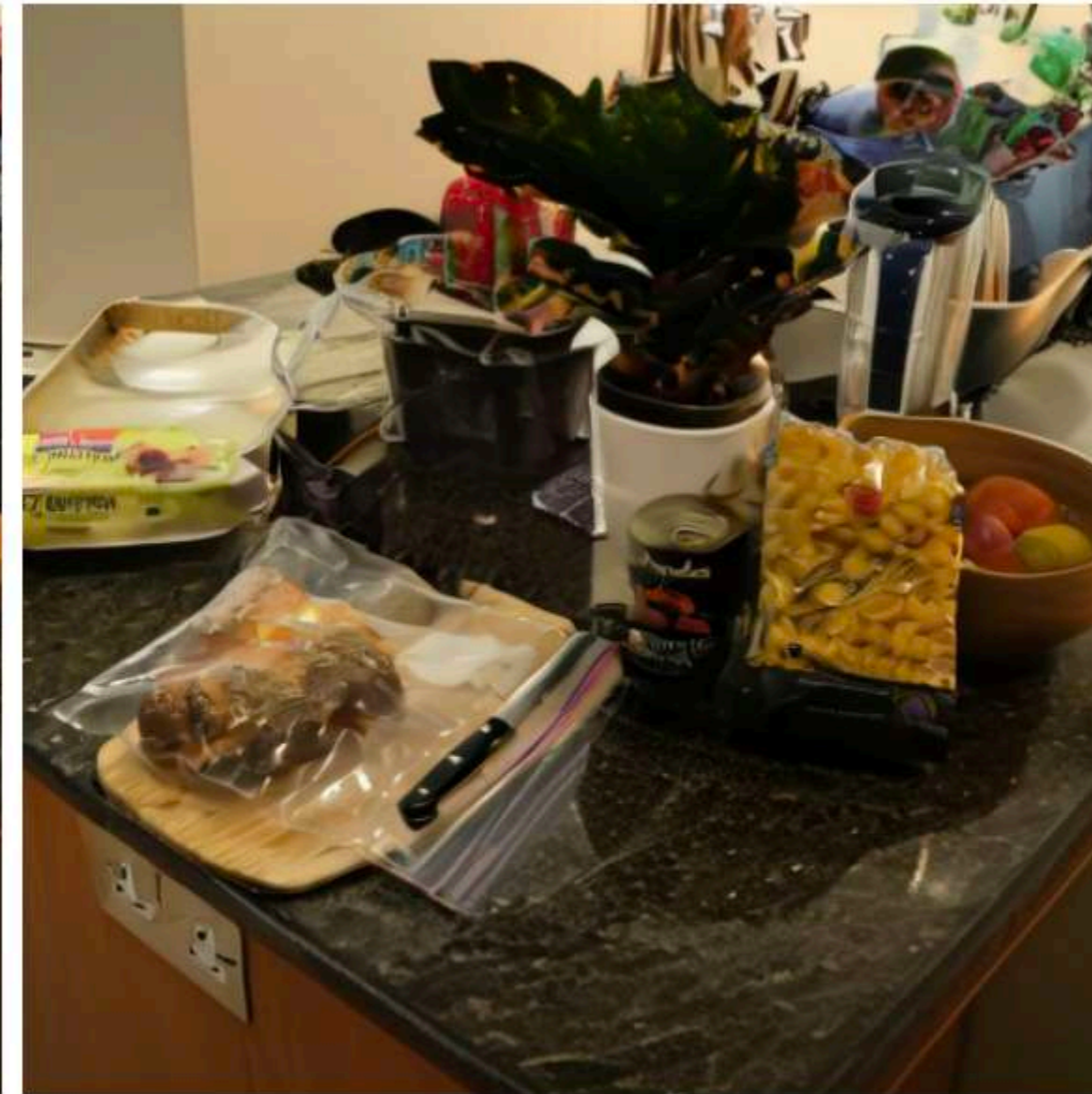
Conditional Generation Examples



(a) 3 input views



(b) 6 input views



(c) 9 input views

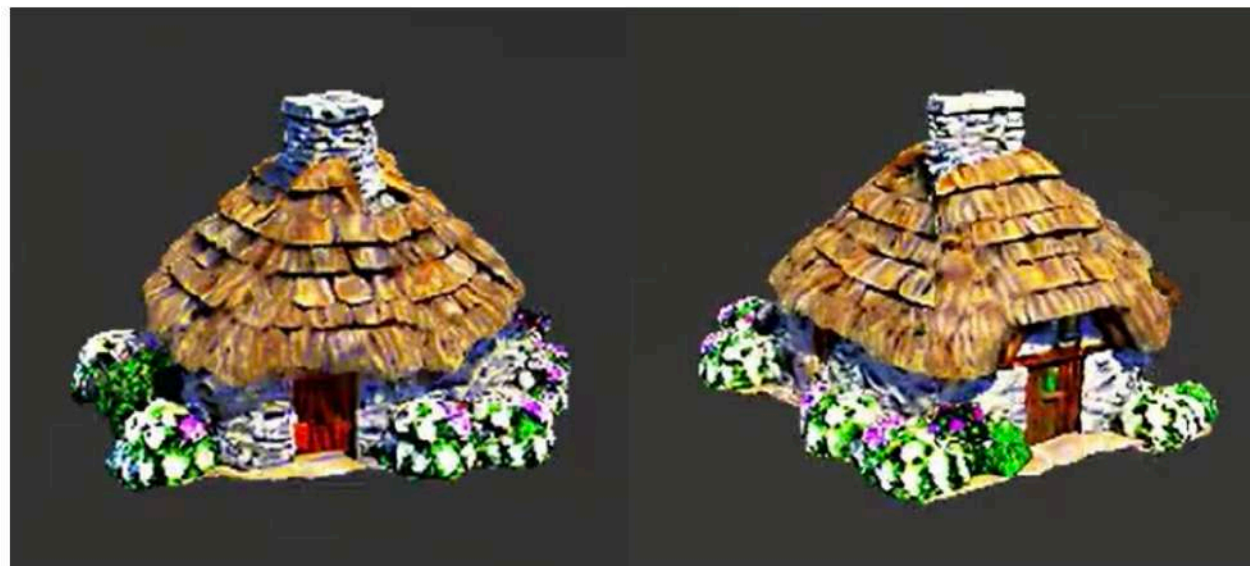
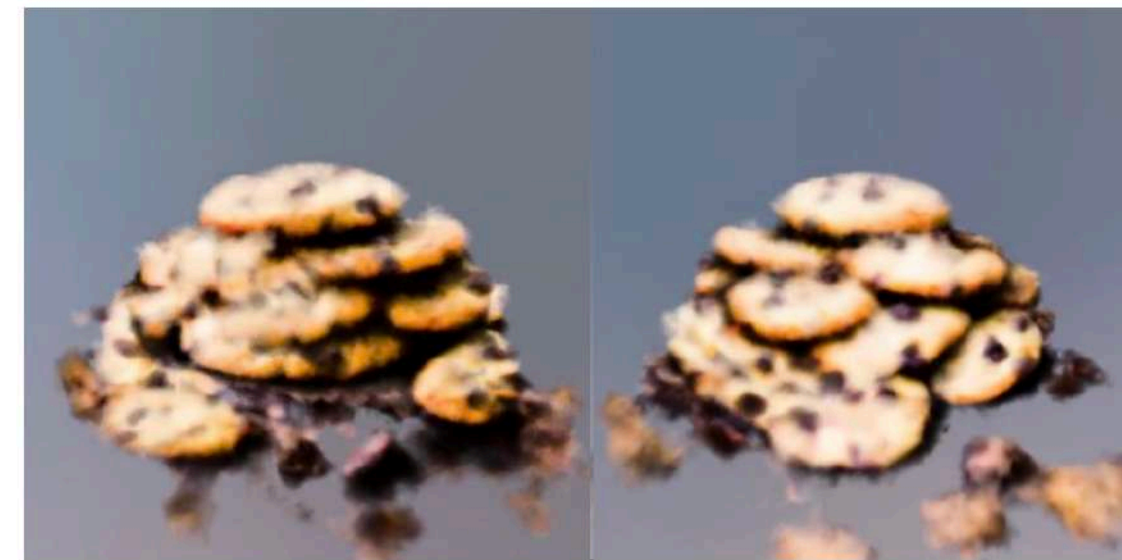
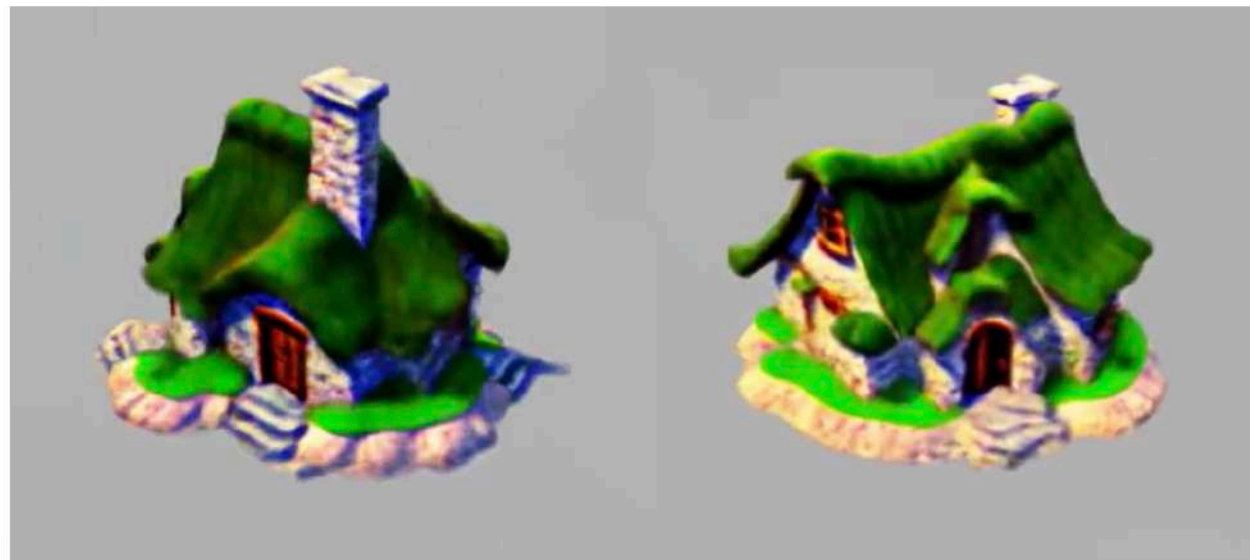


(d) Ground truth

Cross-View Consistency is Still an Issue

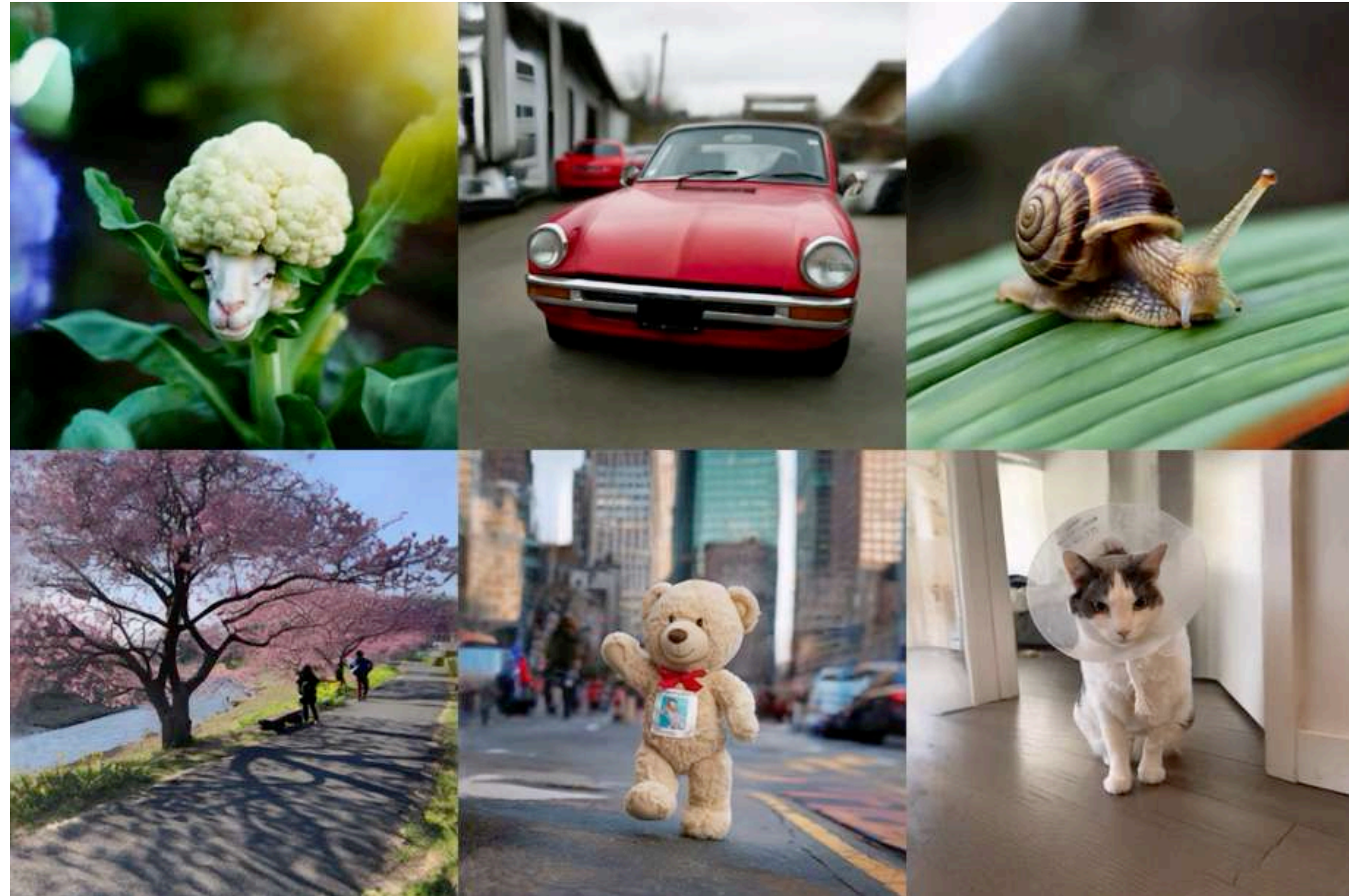


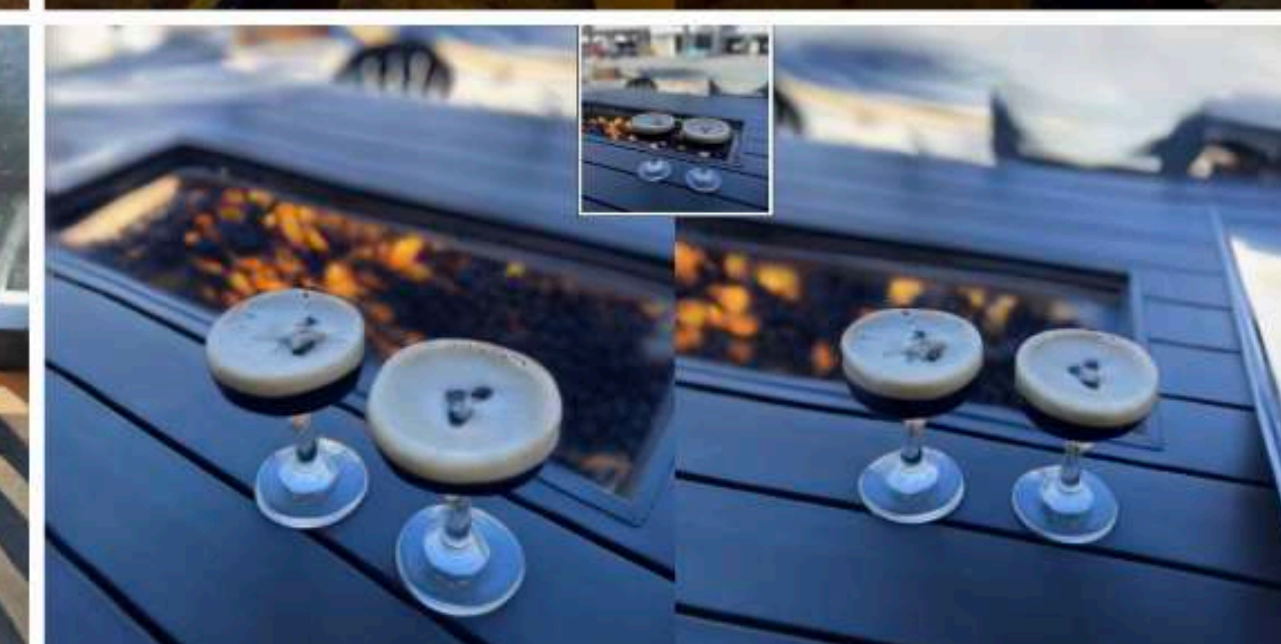
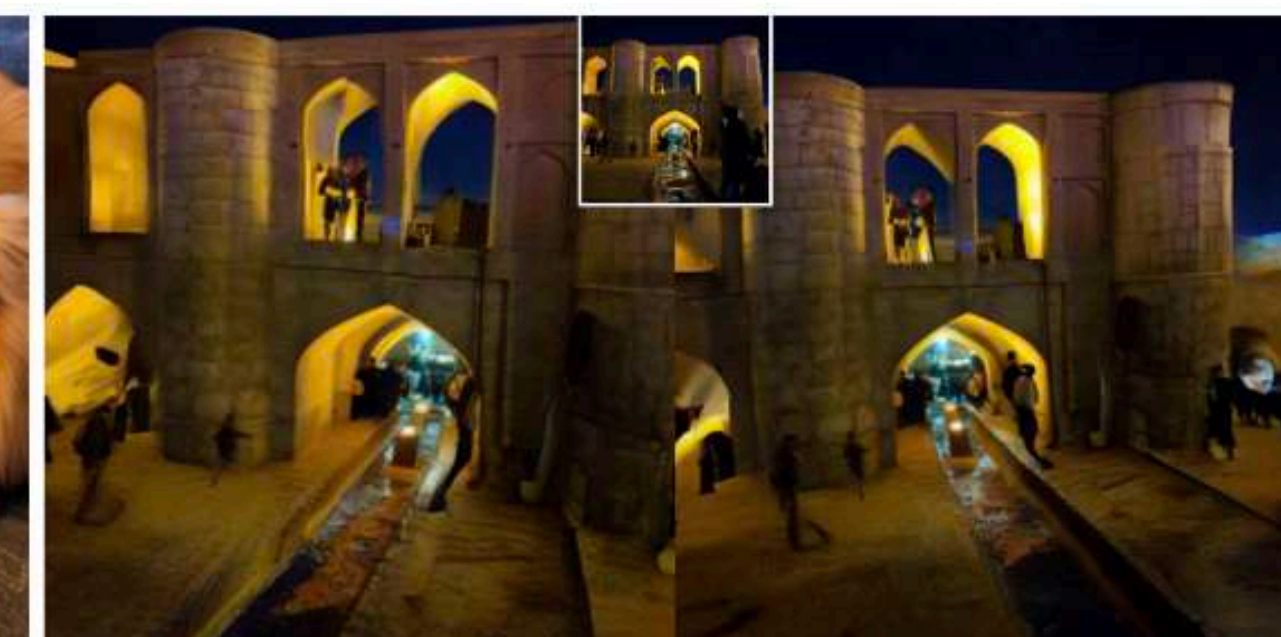
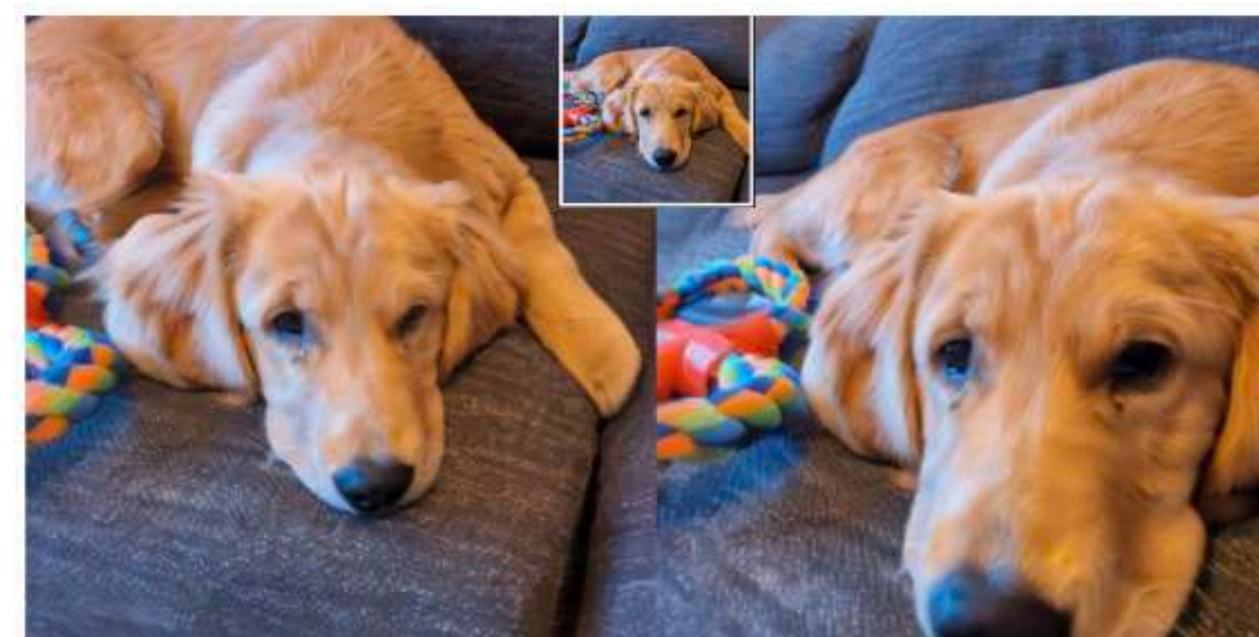
Sora Generates Videos with Stunning Geometrical Consistency



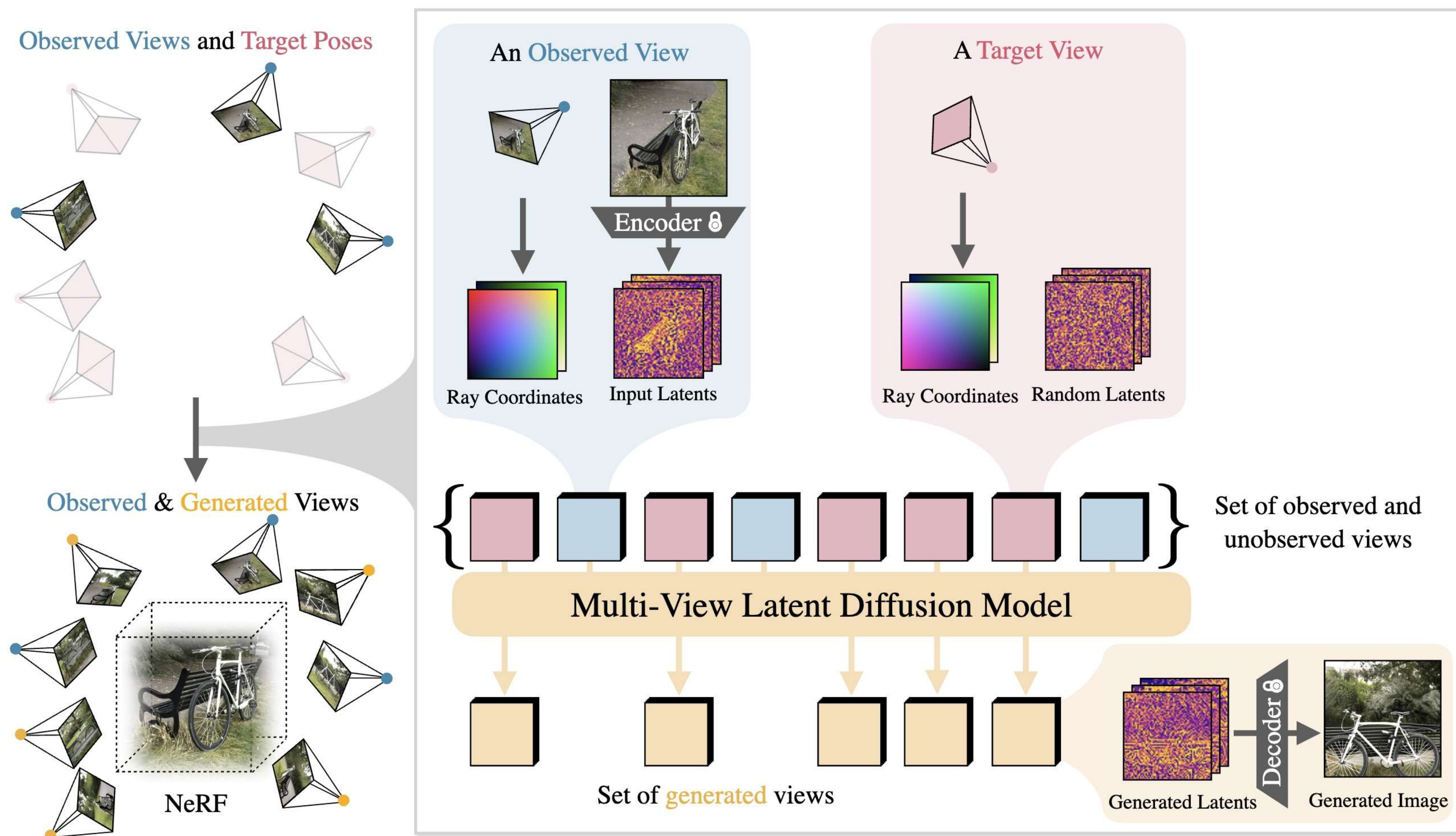
Cat3D

- Novelty
 - Simpler generation pipeline
 - Simpler diffusion conditioning
- Not novel, yet important:
 - Simultaneous generation of views
- Tasks
 - Text-to-3d, Image-to-3D, Novel View Synthesis



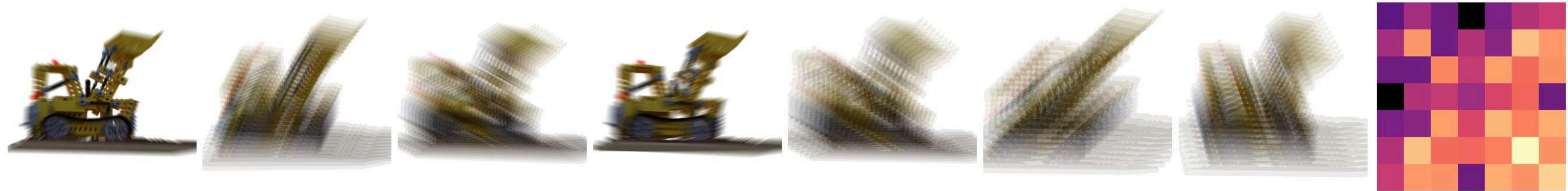


Overall Scheme for Data Generation



A Sidenote of Camera Embeddings

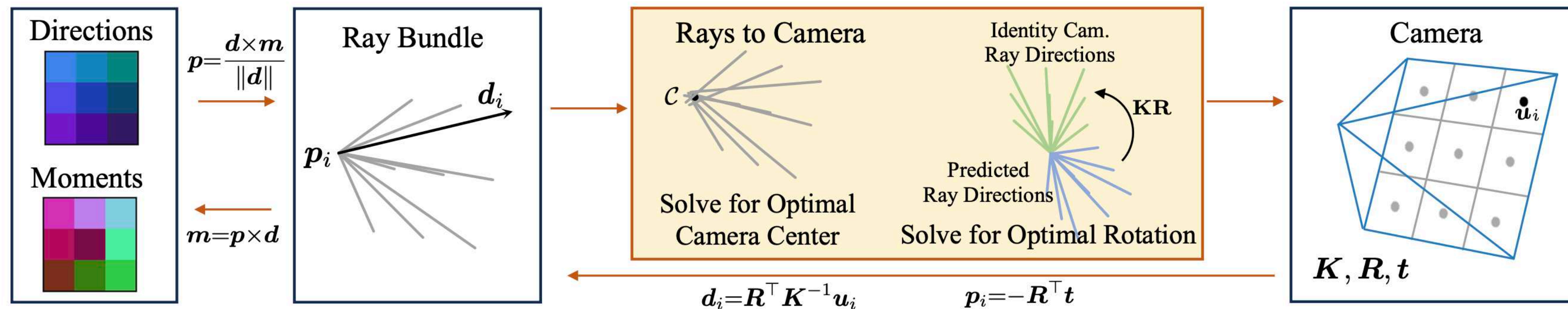
- A naive approach is to embed location, rotation and focal length



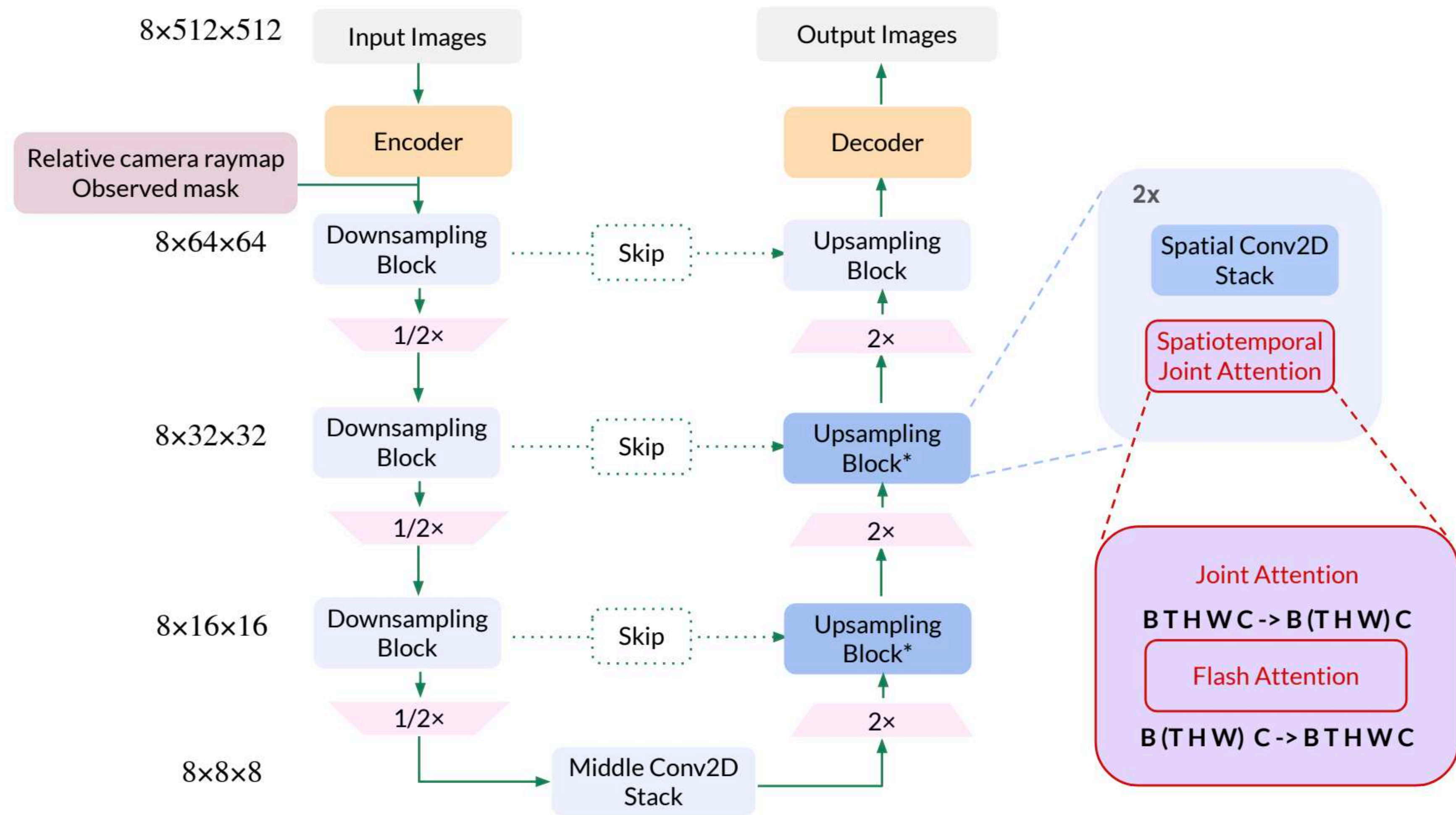
(a) Motion trails for **SE3+Focal**

(b) Covariance

- Nowadays, the go-to approach is to represent camera with a bundle of rays

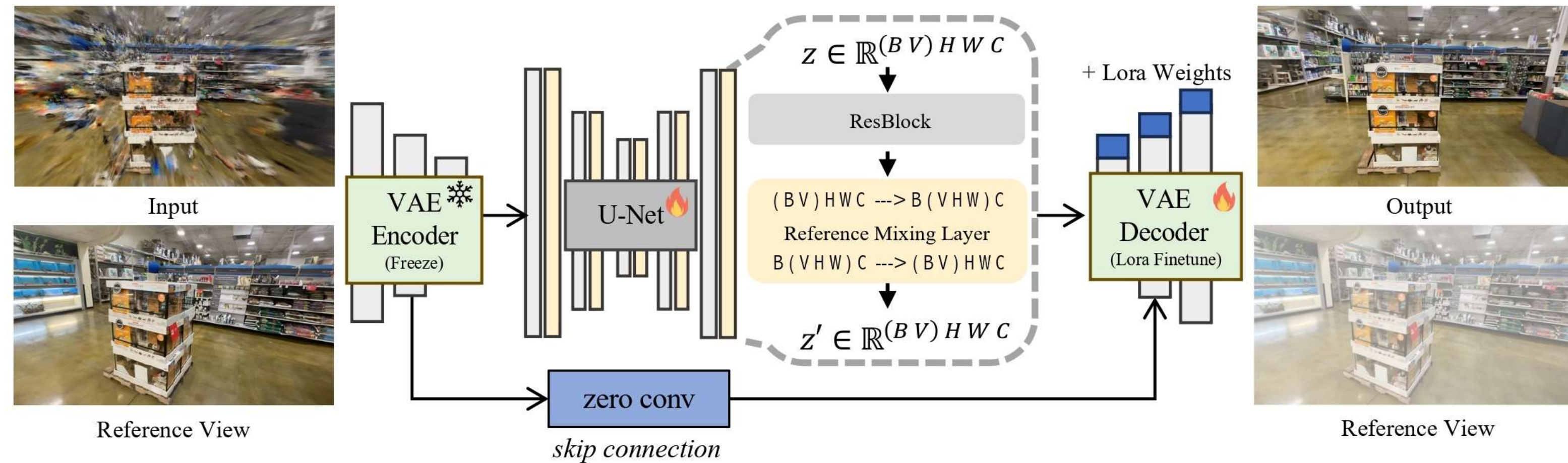


Diffusion Model Architecture



Difix3D

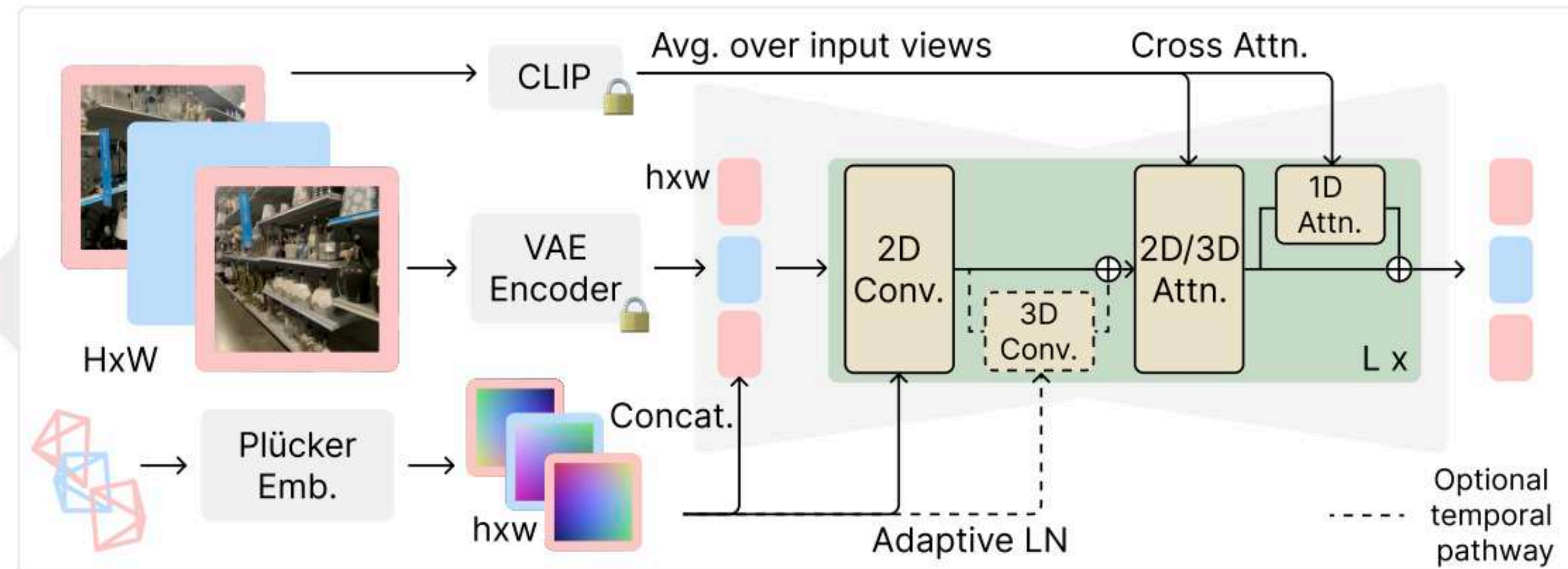
Fix Splats Instead of Generating Data from Scratch



Stable Virtual Camera

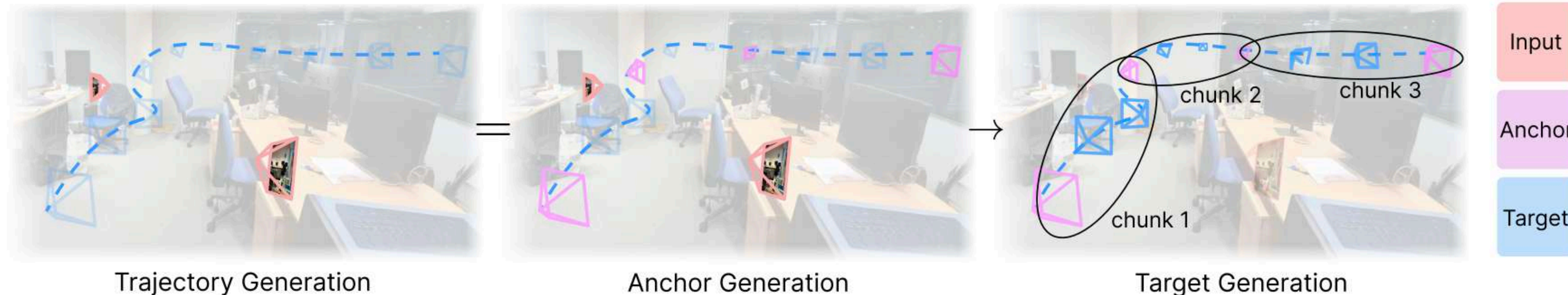
Temporal Consistency, No 3D Distillation, Open Source

Training: fixed seq. len (M -in N -out)



Sampling: variable seq. len (P -in Q -out)

Procedural Two-Pass Sampling



Stable Virtual Camera

Single frame



Multiple Frames:

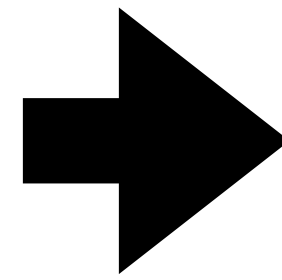


World Models

The Fundamental Limitation of 3D

- Applications so far assumed a static scene captured from multiple viewpoints
 - Expensive to collect, fragile
- Meanwhile, video data is abundant and captures the world in all its complexity
- Scaling simple models on abundant data sooner or later beats methods relying on prior knowledge in the data (i.e., static world, calibrated camera)

The Component Video Diffusion is Missing



interactivity

Interactivity

- Why?
 - It's fun!
 - A sandbox for RL agents to operate in (aka *world models*)
- How?
 - Autoregressive inference
 - Responsive to user inputs

From Video to World Models

A Proprietary Stack We Can Only Speculate On



Veo



Genie 3



Sima 2

<https://deepmind.google/models/veo/>

<https://deepmind.google/models/genie/>

<https://deepmind.google/blog/sima-2-an-agent-that-plays-reasons-and-learns-with-you-in-virtual-3d-worlds/>

My *Speculations*

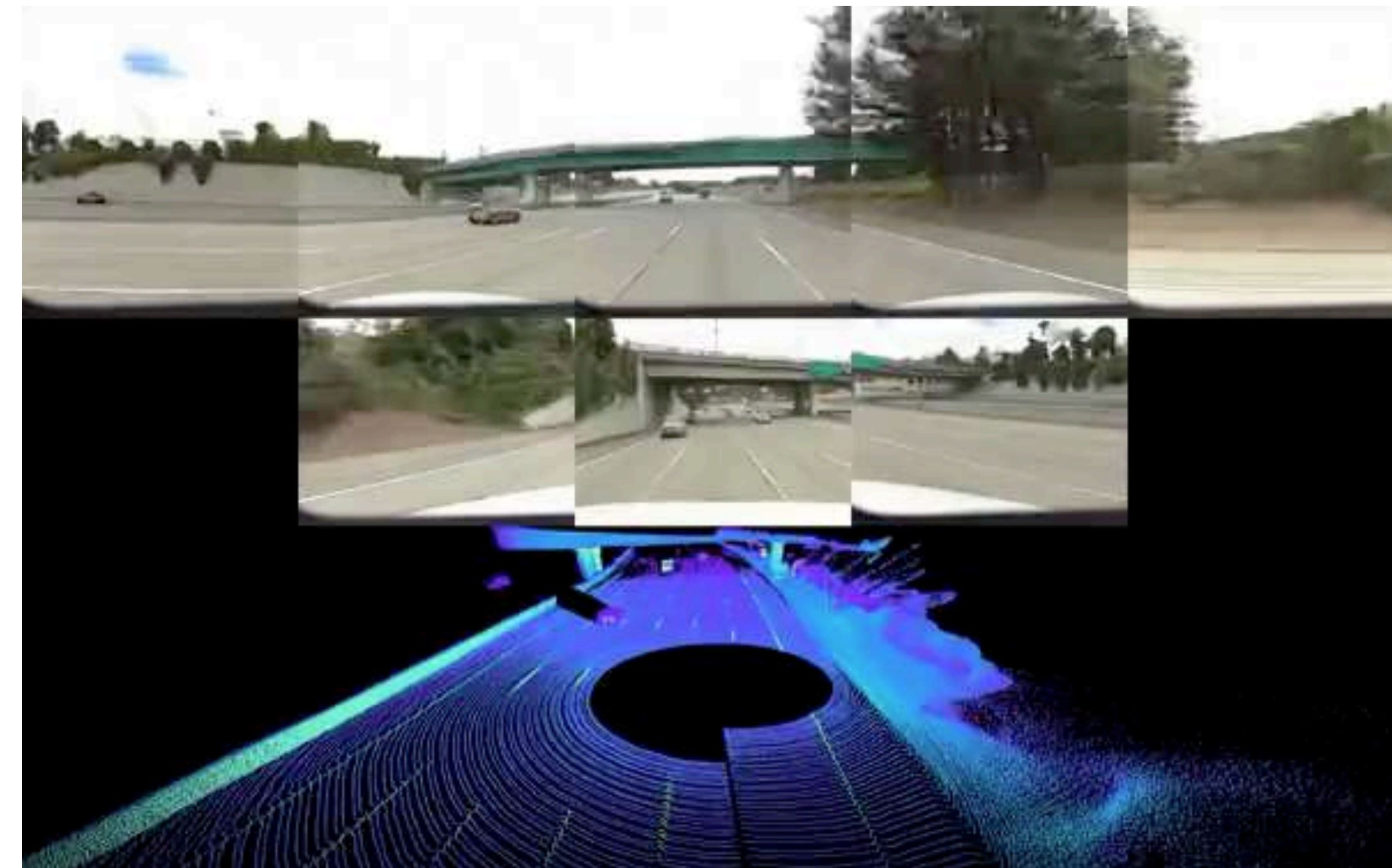
On the Proprietary Stack We Can Only Speculate On

- The model generates a navigable world based on a text (+ image) prompt(s)
- *Training adapts a video generation model to additional conditions*
 - Keystrokes & mouse movements convert into **camera conditioning**
 - **VLA model** describes the events during generation
 - 3D consistency emerges from data, **no explicit geometry**
- Data is a mix of computer games & *labeled preprocessed videos*
- Heavy distillation to achieve **autoregressive generation** at 24 FPS / 720p

World Models for Self-Driving

Sandbox for Robots

- Popular domain with several nuances
 - Multiple output views + LiDAR simulation
 - Modelling other agents is a crucial
- Competitive field with numerous contributors
 - NVIDIA, Waymo, Wayve, Comma ai, etc.
 - Yandex & Navio also work on the topic



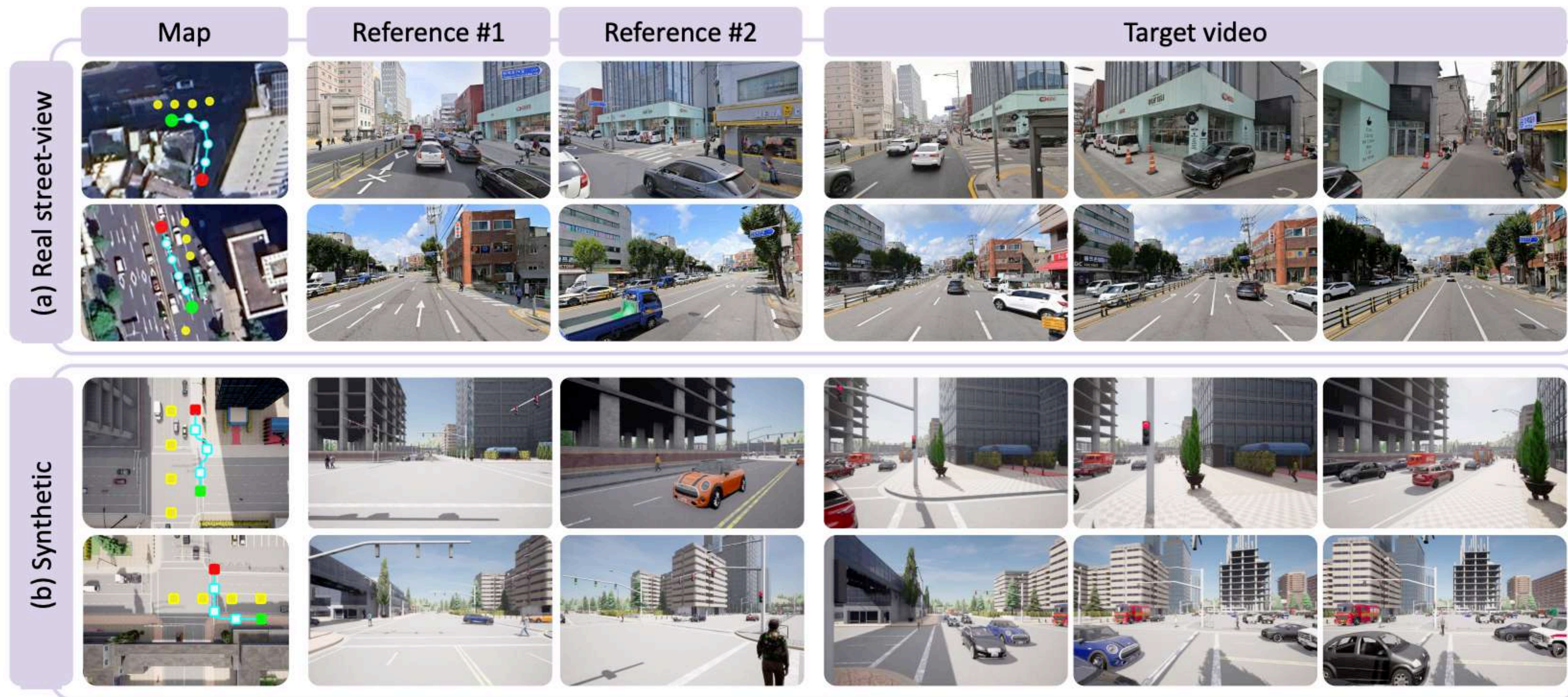
The Role of 3D in the Age of World Models

- If video models can already generate worlds, do we still need explicit 3D?
 - Helpful for training (e.g. video games as data)
 - **3D representations as an interface:** instead of generating a separate stream per user, generate a 3D representation once (see World Labs, Lyra 2)
 - 3D priors helps with memory and consistency: **retrieval-augmented generation** with 3D data



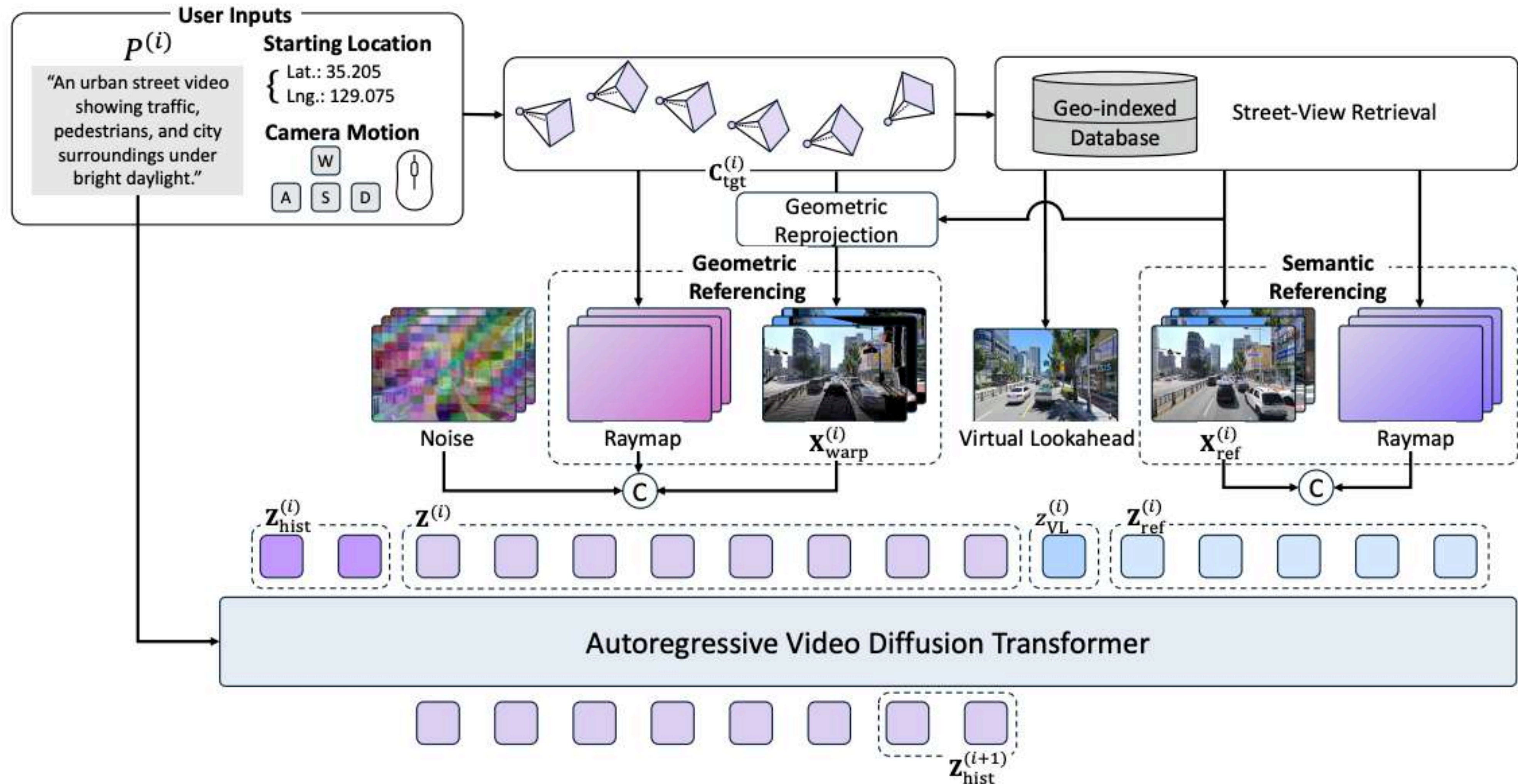
3D Priors in Seoul World Models

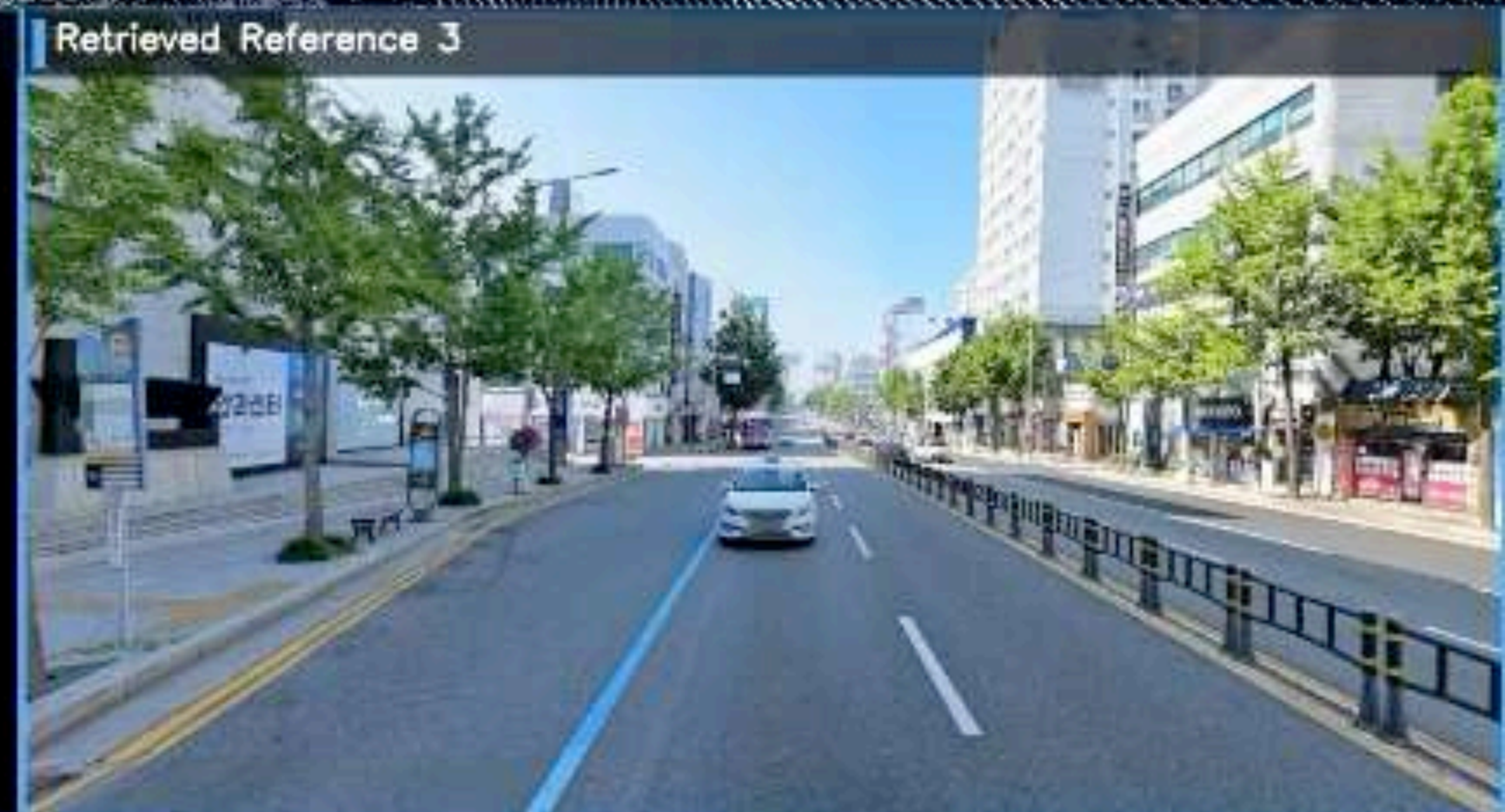
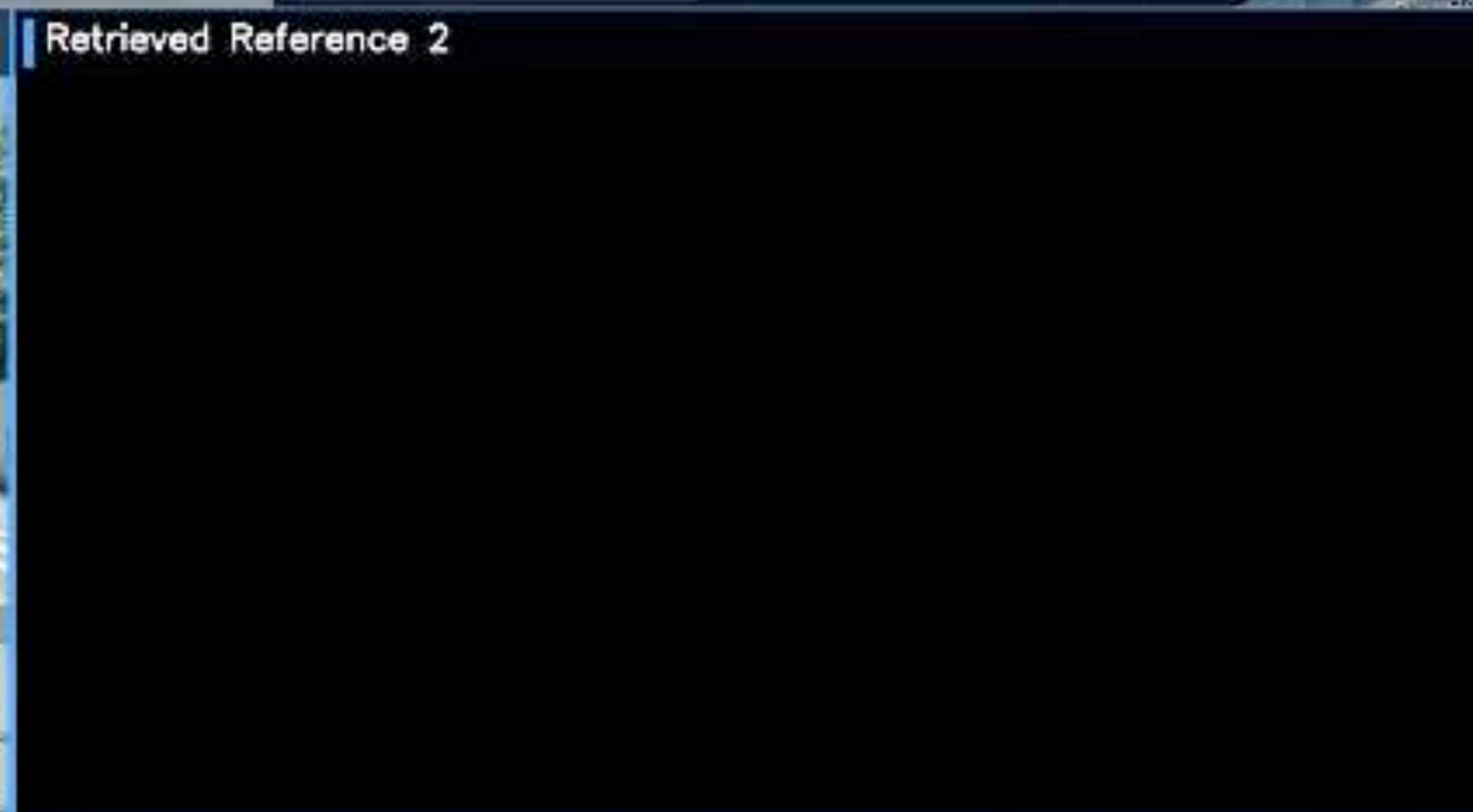
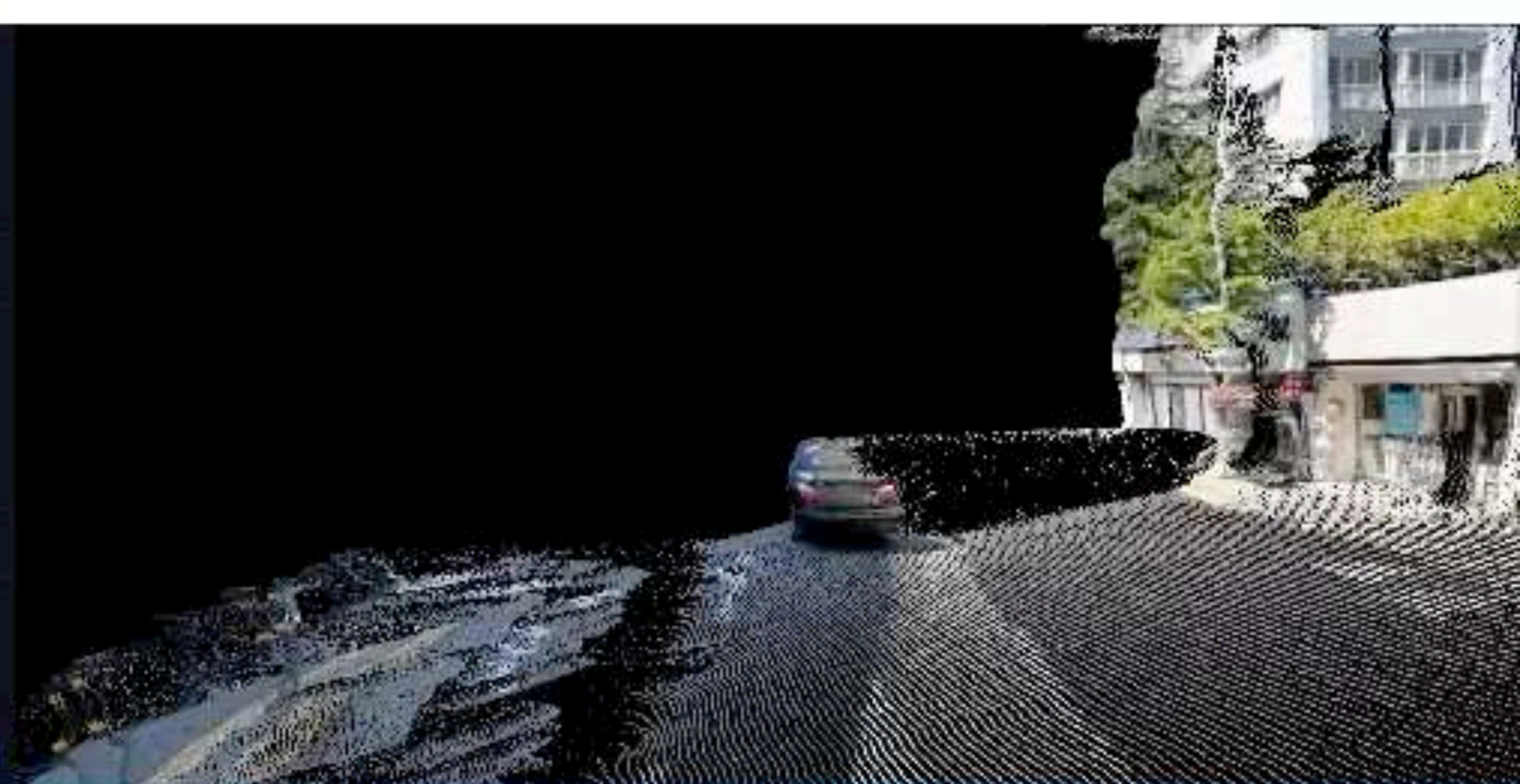
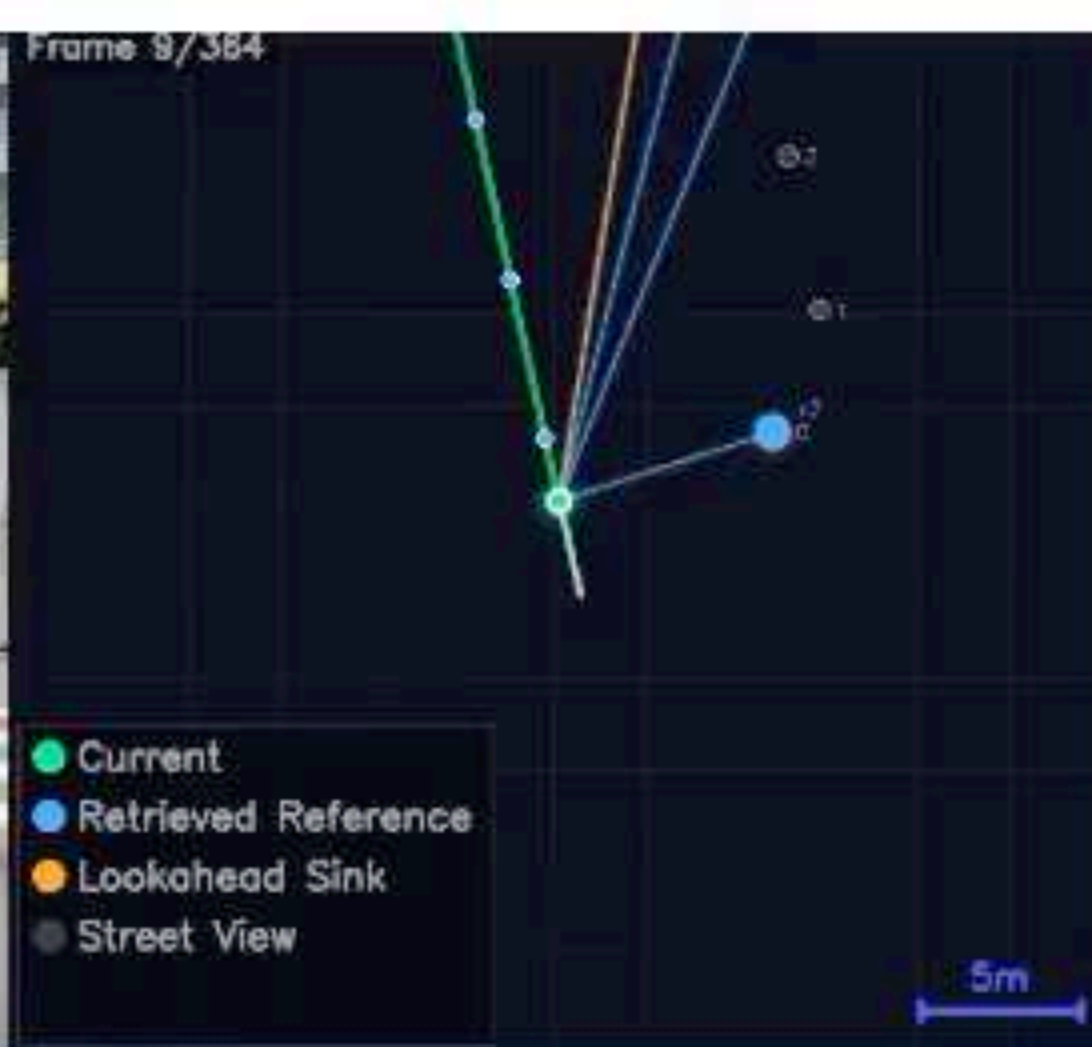
Building City-Scale Driving Simulation with Retrieval-Augmented Generation



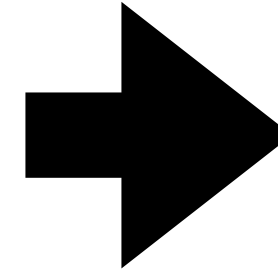
3D Priors in Seoul World Models

Building City-Scale Driving Simulation with Retrieval-Augmented Generation





Summary



2x speed

What if it were **twilight** on this street?

- Pure 3D data is scarce
- As a result, scaling generative models in 3D is challenging and fun
- Diffusion models can be leveraged for 3D generation
 - Score Distillation Sampling (SDS) for direct optimization
 - Fine-tuning 2D priors (the practical current approach)
 - Domain-specific data: 3D for text-to-3D, multi-view for reconstruction
- Video & world models take center stage, will 3D priors be relevant in a year?